

Spinal cord: internal organization

The spinal cord provides innervation for the trunk and limbs via spinal nerves and their peripheral ramifications. It receives primary afferent fibres from peripheral receptors located in widespread somatic and visceral structures, and sends motor axons to skeletal muscle. It also contains the cell bodies of all the preganglionic neurones responsible for the sympathetic innervation of cardiac and smooth muscle and secretory glands, and for the parasympathetic innervation of smooth muscle in the distal part of the hindgut, the pelvic viscera and the erectile tissues of the external genitalia. Many bodily functions are regulated at an unconscious level by intraspinal reflex connections between afferent and efferent neurones. Profuse ascending and descending pathways link the spinal cord with the brain, allowing higher centres to monitor and perceive external and internal stimuli and to modulate and control spinal efferent activity.

EXTERNAL FEATURES AND RELATIONS

The topographical anatomy of the spinal cord and its external features and relations are described in more detail in [Chapter 45](#). In brief, the cord lies within the vertebral canal. It is continuous rostrally with the medulla oblongata, just below the level of the foramen magnum and it terminates caudally as the conus medullaris, which is continuous with the filum terminale and is anchored to the dorsum of the coccyx. The cord is ensheathed by spinal meninges that are continuous with the cranial meninges through the foramen magnum. Although it is approximately circular in cross-section, the diameter of the spinal cord varies according to level; it bears two enlargements, cervical and lumbar.

The spinal cord is essentially a segmental structure, giving rise to 31 bilaterally paired spinal nerves. These attach to the cord as a linear series of smaller dorsal and ventral nerve rootlets. Dorsal rootlets contain afferent nerve fibres and ventral rootlets contain efferent fibres (see [Fig. 16.2](#)). Groups of adjacent rootlets coalesce to form dorsal or ventral nerve roots that cross the subarachnoid space and unite to form functionally mixed spinal nerves as they pass through the intervertebral foramina. The dorsal roots bear dorsal root ganglia that contain the cell bodies of primary afferent neurones.

INTERNAL ORGANIZATION

In transverse section, the spinal cord is incompletely divided into symmetrical halves by a dorsal (posterior) median septum and a ventral (anterior) median fissure (sulcus) ([Fig. 20.1](#)). It consists of an outer layer of white matter and an inner core of grey matter; their relative sizes and configuration vary according to level. The amount of grey matter reflects the number of neuronal cell bodies present; it is proportionately largest in the cervical and lumbar enlargements, which contain the neurones that innervate the limbs. The absolute amount of white matter is greatest at cervical levels and decreases progressively at lower levels because descending tracts shed fibres as they descend and ascending tracts accumulate fibres as they ascend.

A diminutive central canal, lined by columnar, ciliated epithelium (ependyma) and containing cerebrospinal fluid (CSF), extends the whole length of the spinal cord, lying in the centre of the spinal grey matter. Rostrally, the central canal extends into the caudal half of the medulla oblongata and then opens into the fourth ventricle.

SPINAL GREY MATTER

In transverse section, the spinal grey matter is often described as being 'butterfly-shaped' or resembling the letter 'H' (see [Fig. 20.1](#)). It consists

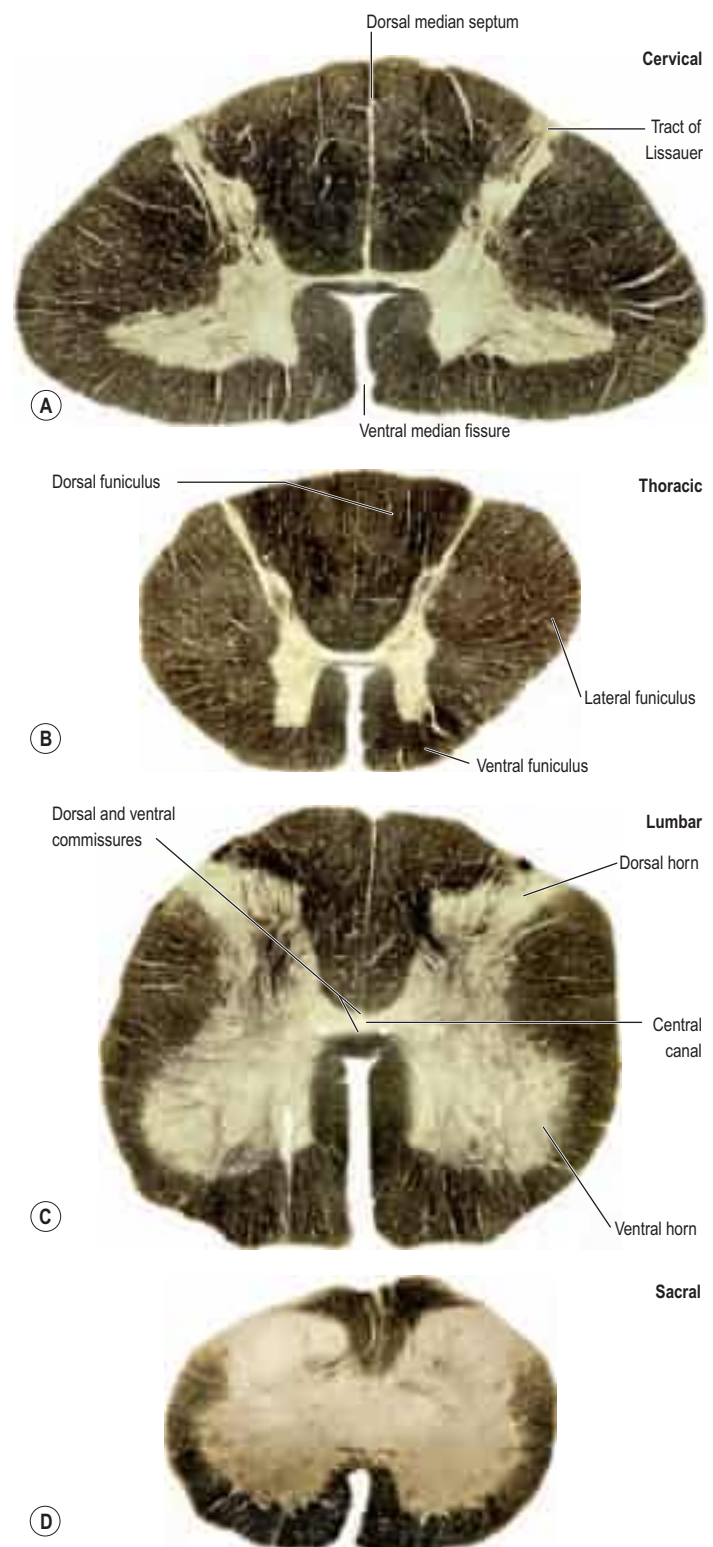


Fig. 20.1 Transverse sections through the spinal cord at representative levels. Approximately $\times 5$. (Figure enhanced by B Crossman.)

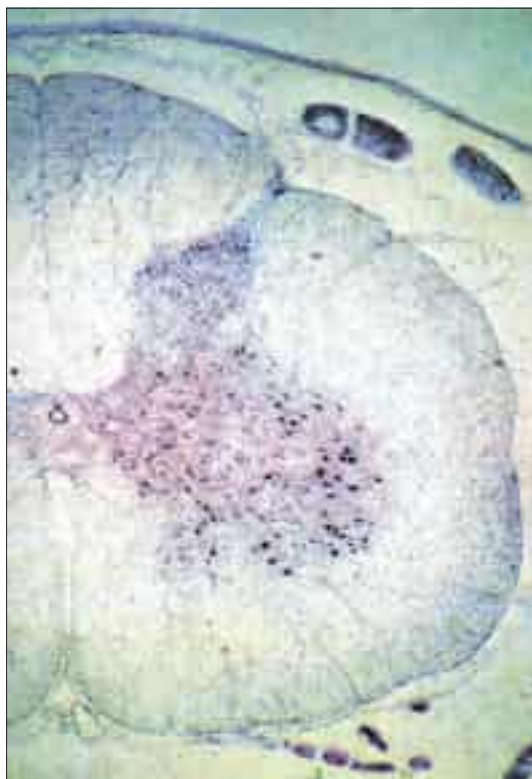


Fig. 20.2 A transverse section of a spinal cord at a mid-lumbar level. Groups of the larger motor neurones in the ventral grey column are prominent. Stained with cresyl fast violet.

of four linked cellular masses: the right and left dorsal horns, which project dorsolaterally, and the right and left ventral horns which project ventrolaterally towards the surface. The grey matter that immediately surrounds the central canal and unites the two sides constitutes the dorsal and ventral grey commissures. The dorsal horn is the site of termination of the primary afferent fibres that enter the cord via the dorsal roots of spinal nerves. The tip of the dorsal horn is separated from the dorsolateral surface of the cord by a thin fasciculus or tract (of Lissauer), in which primary afferent fibres ascend and descend for a short distance before terminating in the subjacent grey matter. The ventral horn contains efferent neurones whose axons leave the spinal cord in ventral nerve roots. A small intermediate, or lateral, horn is present at thoracic and upper lumbar levels; it contains the cell bodies of preganglionic sympathetic neurones.

Spinal grey matter (**Fig. 20.2**) is a complex mixture of neuronal cell bodies, their processes and synaptic connections, neuroglia and blood vessels. Neurones in the grey matter are multipolar. They vary in size and features such as the length and the arrangement of their axons and dendrites. Neurones may be intrasegmental, i.e. contained within a single segment, or intersegmental, i.e. their ramifications spread through several segments.

Neuronal cell groups of the spinal cord

The grey matter of the spinal cord may be divided, on the basis of cytoarchitecture, into 10 zones, known as Rexed's laminae, which are numbered sequentially from dorsal to ventral. The laminae are defined on the basis of neuronal size, shape, cytological features and density. Some of these lamina are equated with cell groupings of particular functional types (**Fig. 20.3**) (Rexed 1952).

Laminae I–IV correspond to the dorsal part of the dorsal horn, and are the main site of termination of cutaneous primary afferent terminals and their collaterals. Many complex polysynaptic reflex paths (ipsilateral, contralateral, intrasegmental and intersegmental) start from this region, as also do many long ascending tract fibres, which pass to higher levels. Lamina I (lamina marginalis) is a very thin layer with an ill-defined boundary at the dorsolateral tip of the dorsal horn. It has a reticular appearance, reflecting its content of intermingling bundles of coarse and fine nerve fibres. It contains small, intermediate and large neuronal somata, many of which are fusiform in shape. The much larger lamina II consists of densely packed small neurones, responsible for its dark appearance in Nissl-stained sections. With myelin stains, lamina II is characteristically distinguished from adjacent laminae by

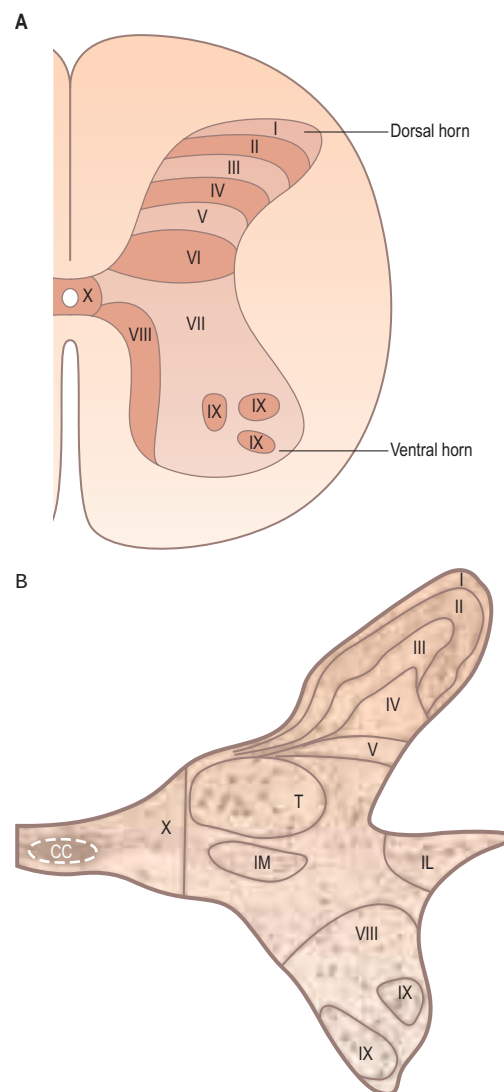


Fig. 20.3 Rexed's laminae shown in the cervical cord (**A**) and in the thoracic cord (**B**). Abbreviations: CC, central canal; IL, intermediolateral; IM, intermediomedial; T, posterior thoracic nucleus. (A, Redrawn with permission from Crossman AR, Neary D 2010 *Neuroanatomy*, 4th edn. Edinburgh: Churchill Livingstone; B, Redrawn with permission from Carpenter MB 1991 *Core Text of Neuroanatomy*, 4th edn. Baltimore: Williams and Wilkins.)

the almost total lack of myelinated fibres. Lamina II corresponds approximately to the substantia gelatinosa. Lamina III consists of somata that are mostly larger, more variable and less closely packed than those in lamina II. It also contains many myelinated fibres. The ill-defined nucleus proprius of the dorsal horn corresponds to some of the cell constituents of laminae III and IV. Lamina IV is a thick, loosely packed, heterogeneous zone permeated by fibres. Its neuronal somata vary considerably in size and shape, from small and round, through intermediate and triangular, to very large and stellate (Schoenen and Faull 1990).

Laminae V and VI lie at the base of the dorsal horn. They receive most of the terminals of proprioceptive primary afferents, profuse corticospinal projections from the motor and sensory cortex, and input from subcortical levels, suggesting their involvement in the regulation of movement. Lamina V is a thick layer, divisible into a lateral third and medial two-thirds. Both have a mixed cell population but the former contains many prominent well-staining somata interlaced by numerous bundles of transverse, dorsoventral and longitudinal fibres. Lamina VI is most prominent in the limb enlargements. It has a densely staining medial third of small, densely packed neurones and a lateral two-thirds containing larger, more loosely packed, triangular or stellate somata.

Lamina VII occupies the region just ventral to lamina VI and extending across the spinal grey matter on each side. This region is known as the intermediate zone and within the thoracic cord includes the lateral horn. The size of lamina VII varies at different spinal levels. In the cervical and lumbar enlargements, lamina VII extends laterally and ventrally

throughout the ventral horn containing cell groups of lamina IX neurones embedded within it. In the thoracic region, lamina VII occupies the intermediate zone and the dorsal part of the ventral horn only. Three important nuclear groups are contained within lamina VII: the posterior thoracic nucleus (nucleus thoracicus posterior, Clarke's column), which extends throughout the thoracic and upper lumbar segments and gives rise to the dorsal spinocerebellar tract; the intermediolateral nucleus, which is located between T1 and L2 and gives rise

to the preganglionic sympathetic fibres; and the intermediomedial nucleus, which extends the full length of the cord and may be involved in the control of visceral motor neurones (Fig. 20.4).

Lamina VIII spans the base of the ventral horn in the thoracic region but is restricted to the medial aspect of the ventral horn in the cervical and lumbar enlargements. Its neurones display a heterogeneous mixture of sizes and shapes from small to moderately large. Lamina VIII is a mass of propriospinal interneurons. It receives terminals from the

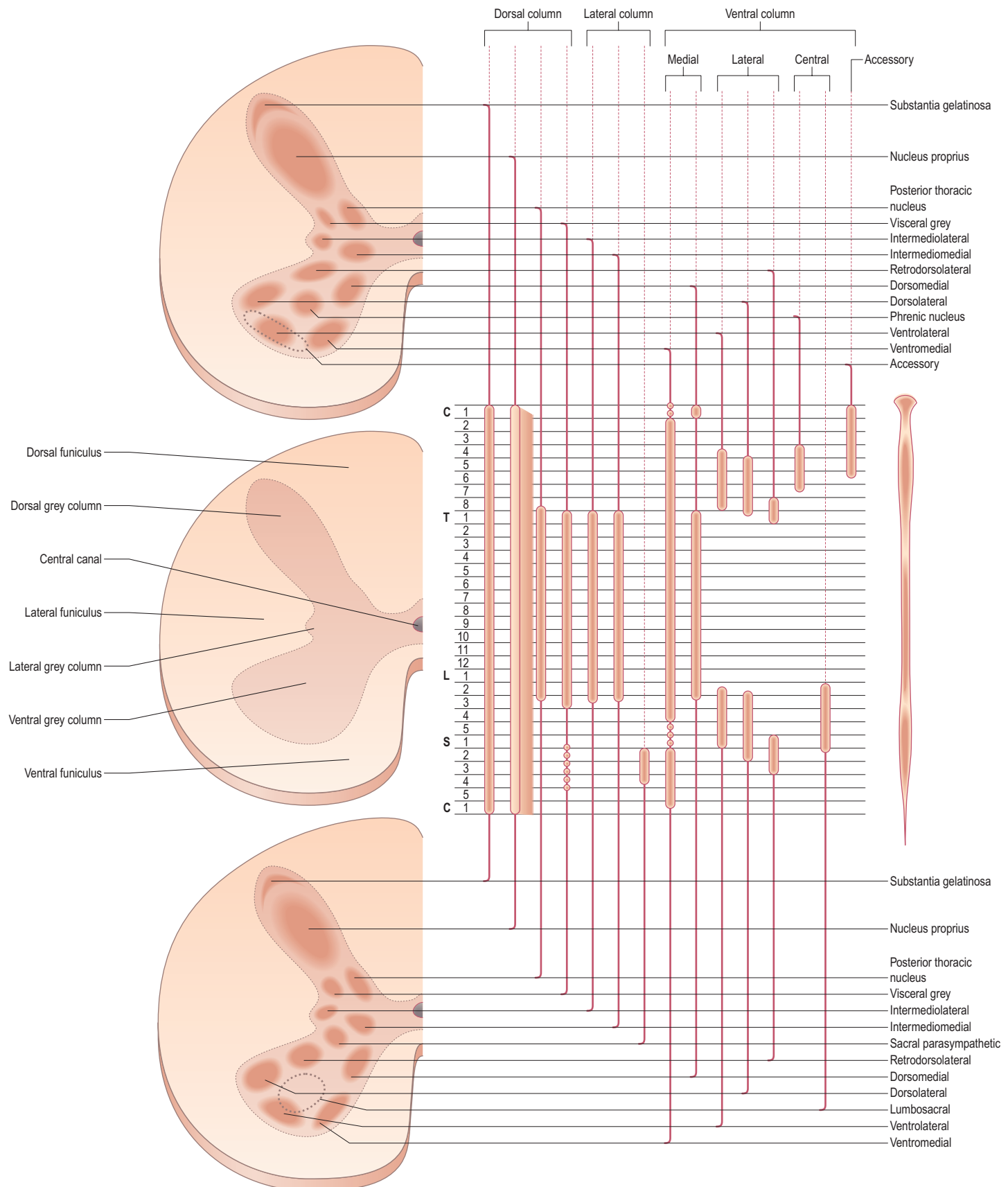


Fig. 20.4 The groups of nerve cells in the grey columns of the spinal cord. The relative positions of these columnar groups and their extent through spinal segments are indicated. (Modified with permission of Simon & Schuster from *Correlative Anatomy of the Nervous System* by E Crosby, T Humphrey, E Lauer. Copyright © 1962 Macmillan Publishing Company.)

adjacent laminae, many commissural fibres from the contralateral lamina VIII, and descending connections from the interstitiospinal, reticulospinal and vestibulospinal tracts and the medial longitudinal fasciculus. The axons from these interneurons influence α motor neurone activity bilaterally, perhaps directly but more probably by excitation of small γ motor neurones supplying efferent fibres to muscle spindles.

Lamina IX consists of several distinct groups of somatic motor neurones located throughout the ventral horn. In the thoracic cord, these nuclear groups are embedded within lamina VIII, whereas in the cervical and lumbar enlargements, the lamina IX cell groups are embedded within both lamina VII and lamina VIII. Lamina IX is a complex array of cells consisting of α and γ motor neurones and many interneurons. The large α motor neurones supply motor end-plates of extrafusal muscle fibres in striated muscle. Recording techniques have demonstrated tonic and phasic α motor neurones. The former have a lower rate of firing and lower conduction velocity, and tend to innervate type S muscle units. The latter have higher conduction velocity and tend to supply fast twitch (type FR, FF) muscle units. The smaller γ motor neurones give rise to small-diameter efferent axons (fusimotor fibres), which innervate the intrafusal muscle fibres in muscle spindles. There are several functionally distinct types of γ motor neurone. The 'static' and 'dynamic' responses of muscle spindles have separate controls mediated by static and dynamic fusimotor fibres, which are distributed variously to nuclear chain and nuclear bag fibres (Boyd and Gladden 1985).

Lamina X surrounds the central canal and consists of the dorsal and ventral grey commissures.

Dorsal horn

The dorsal horn is a major zone of termination of primary afferent fibres, which enter the spinal cord through the dorsal roots of spinal nerves. Dorsal root fibres contain numerous molecules, which are either known, or suspected, to fulfil a neurotransmitter or neuromodulator role. These include glutamic acid, substance P, calcitonin gene-related peptide (CGRP), bombesin, vasoactive intestinal polypeptide (VIP), cholecystikinin (CCK), somatostatin, dynorphin and angiotensin II. Dorsal root afferents carry exteroceptive, proprioceptive and interoceptive information. Laminae I–IV are the main cutaneous receptive areas; lamina V receives fine afferents from the skin, muscle and viscera; and lamina VI receives proprioceptive and some cutaneous afferents. Most, if not all, primary afferent fibres divide into ascending and descending branches on entering the cord. These then travel for variable distances in the tract of Lissauer, near the surface of the cord, and send collaterals into the subjacent grey matter.

The lamina marginalis (lamina I) is a thin lamina of neurones at the dorsolateral tip of the dorsal horn, deep to the tract of Lissauer. Beneath it lies the substantia gelatinosa (laminae II), which is present at all levels, and consists mostly of small neurones, together with some larger neurones. The substantia gelatinosa receives afferents via the dorsal roots, and its neurones give rise to fibres that form the contralateral spinothalamic tract. The large propriospinal neurones of the nucleus proprius lie ventral to the substantia gelatinosa within lamina III and lamina IV; they link segments for the mediation of intraspinal coordination (see Fig. 20.2).

The posterior thoracic nucleus lies at the base of the dorsal horn within lamina VII. At most levels, it is near the dorsal white funiculus and may project into it. It can usually be identified from the eighth cervical to the third or fourth lumbar segments. Neurones of the posterior thoracic nucleus vary in size but most are large, especially in the lower thoracic and lumbar segments. Some send axons into the dorsal spinocerebellar tracts and others are interneurons.

Pain pathways

Fifty years ago, a paper was published that not only had a profound effect on thinking about sensory processing and perception at the time but also continues to influence research in the field of pain perception. The authors described a mechanism by which nociceptive input from the skin was modulated ('gated') by local circuits within the dorsal horn and by descending inputs from the brain; the theory was known as the gate control theory (GCT; Melzack and Wall 1965). Interneurons in the substantia gelatinosa, SG cells, were cast in the role of the metaphorical gate, modulating patterns of afferent input to noci-responsive projection neurones (T cells, tract cells), found mainly in lamina I of the dorsal horn, and responsible for onward transmission of noxious messages to the brain. According to the GCT, increased activity in the collaterals of large-calibre innocuous afferents excited SG cells, which

in turn inhibited the T cells, i.e. the gate was closed. Activity in small-calibre nociceptive afferents had the opposite effect of inhibiting the SG interneurons and opening the gate (Fig. 20.5). Since all primary afferents are excitatory, the model implied that inhibition of the SG cells was the product of interneuronal circuitry, although the participating cells had yet to be identified.

More is now known about the dorsal horn circuitry that generates the gate. Interneurons in laminae I–IV are morphologically, topographically and neurochemically heterogeneous: islet, central, radial and vertical cells have been identified in experimental animal studies. Islet cells are predominantly GABAergic (transmitting or secreting γ -aminobutyric acid); a subset co-express glycine and are presumed to be exclusively inhibitory. Radial and vertical cells are predominantly glutamatergic and presumed to be excitatory, and central cells include both inhibitory and excitatory subsets. The input from C nociceptors is concentrated in lamina I and lamina II; recent evidence has refocused attention on the concept of labelled lines (each sensory modality is processed via a fixed path from skin to brain) for noxious heat, cold, noxious mechanical stimulation and pruritogens, at least at the level of the primary afferent input. The main elements in the circuitry by which noxious and innocuous primary afferent inputs are thought to influence projection neurones within the dorsal horn of the spinal cord are shown in Figure 20.6. (For further reading, see Ma (2010), Moayedi and Davis (2013), Todd and Koerber (2013), Zhang et al (2013), Braz et al (2014), Mendell (2014).)

Onward transmission in the lateral spinothalamic tract is likely to evoke the perception of pain at supraspinal levels, although the activity of spinothalamic tract neurones may be modified by descending pain modulatory systems. In the brainstem, the regions inducing such effects correspond to a number of midbrain and rhombencephalic nuclei, which, with their connections, constitute an endogenous analgesic

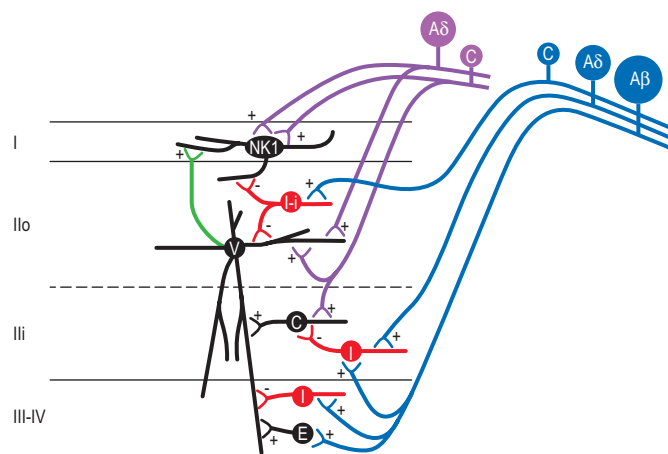


Fig. 20.6 Routes by which noxious and innocuous primary afferent inputs may engage projection neurones in lamina I. Note that the excitatory activation circuits have a ventral-to-dorsal organization. Nociceptive C and A δ fibres (purple) activate vertical cells (V) of the outer part of lamina II (IIo) directly and these, in turn, excite neurokinin 1 (NK1)-expressing projection neurones of lamina I (green axon). Nociceptive C fibres also activate central cells (C) of lamina II (Iii); the C cells contact, and probably excite, V cells in lamina IIo. The projection neurones in lamina I also receive direct nociceptive C and A δ input. Non-nociceptive A β fibres (blue) project to deeper laminae (III–IV), where they establish monosynaptic connections with local excitatory interneurons (E), which directly activate V cells. The majority of the inhibitory neurones (I) in lamina II are islet cells (I–I). These dorsally directed circuits are the route through which both noxious and innocuous primary afferent input can engage the projection neurones of lamina I. The excitatory circuits in the superficial dorsal horn are subject to profound inhibitory controls (red). The majority of the inhibitory neurones (I) in lamina II are islet cells (I–I) and these can be directly engaged by input from low-threshold, mechanoreceptive C fibres (blue). The islet cells in turn establish monosynaptic inhibitory connections with both vertical cells of lamina IIo and NK1 receptor-expressing projection neurones of lamina I. Low-threshold A δ (D-hair; blue) and A β fibres also directly engage inhibitory interneurons in laminae III–IV. The latter in turn exert inhibitory control of a variety of excitatory interneurons, including vertical (V) and central (C) cells. The A β -to-inhibitory cell circuit presumably underlies the circuit through which the 'gate' of the gate control theory may be closed. (Redrawn with permission from Braz J, Solorzano C, Wang X, et al 2014. Transmitting pain and itch messages: a contemporary view of the spinal cord circuits that generate gate control, *Neuron* 82:522–536.)

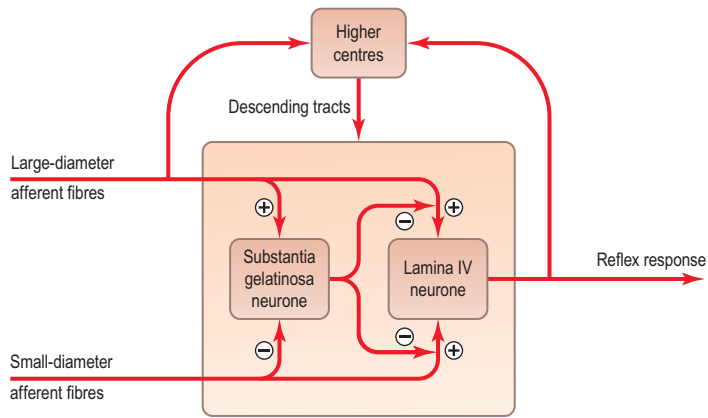


Fig. 20.5 The basic arrangement of the sensory 'gate' mechanism in the dorsal laminae of the grey matter of the spinal cord. (Redrawn with permission from Melzack R, Wall PD. Pain mechanisms: a new theory. *Science* 150:971–979. Copyright 1965 American Association for the Advancement of Science.)

system. In the midbrain, these regions are the periaqueductal grey matter, dorsal raphe nucleus and part of the cuneiform nucleus. Neurons in these sites contain 5-HT (5-hydroxytryptamine, serotonin), GABA, substance P, CCK, neurotensin, enkephalin and dynorphin. The periaqueductal grey matter receives afferents from the frontal somatosensory and cingulate neocortex, the amygdala, numerous local reticular nuclei and the hypothalamus. Afferents from the latter are separate bundles, which carry histamine, luteinizing hormone-releasing hormone (LHRH), vasopressin, oxytocin, adrenocorticotrophic hormone (ACTH), melanocyte-stimulating hormone (γ -MSH), endorphin and angiotensin II. Some fibres descend from the periaqueductal grey matter to rhombencephalic centres, while others pass directly to the spinal cord.

In the rhombencephalon, the nucleus raphe magnus and the medial reticular column constitute an important multineuromediator centre. Neurons in these sites contain 5-HT, substance P, CCK, thyrotrophin-releasing hormone (TRH), enkephalin and dynorphin; some neurons contain two or even three neuromediators. Descending corticospinal fibres pass to the nucleus of the spinal tract of the trigeminal nerve and its continuation, the substantia gelatinosa. The latter extends throughout the length of the cord and contains populations of neurons that express many different neuromediators, e.g. GABA, substance P, neurotensin, enkephalin and dynorphin. There is abundant physiological and pharmacological evidence that all of these regions are intimately concerned with the control of nociceptive (and probably other modality) inputs.

Stimulation of forebrain sites, including the periventricular grey matter, the ventral posterolateral nucleus of the thalamus, and the primary sensory (S1) and posterior parietal cortices, inhibit spinothalamic tract cells. In contrast, some spinothalamic tract cells are excited by stimulation of the medullary reticular formation and the primary motor cortex (the latter effect probably being mediated by the corticospinal tract).

Modern neuroimaging and molecular, genetic and behavioural studies confirm that the experience of pain involves a complex network of responsive brain regions. The somatosensory cortices (S1 and S2) and the insula are thought to encode the sensory features of pain, whereas the prefrontal cortex and components of the limbic system (anterior cingulate cortex, amygdala, ventral tegmental area and nucleus accumbens) mediate the affective and contextual aspects of pain. Mechanisms underlying 'deep' visceral, muscular, neuropathic and persistent, chronic pain remain even less clear than those mediating cutaneous pain (Devor 2013). Patients with chronic pain may have anatomical alterations within the regions involved in the cognitive and emotional modulation of pain, e.g. the dorsolateral and medial prefrontal cortex, anterior cingulate gyrus and insula (Bushnell et al 2013, Denk et al 2014) and dysregulation of the descending pain modulation systems (Ossipov et al 2014).

Lateral horn

The lateral horn is a small lateral projection of grey matter located between the dorsal and ventral horns in lamina VII. It is present from the eighth cervical or first thoracic segment to the second or third lumbar segment. It contains the cell bodies of preganglionic sympathetic neurones. These develop in the embryonic cord dorsolateral to the central canal and migrate laterally, forming the intermediolateral cell columns. Their axons travel via ventral spinal roots and white rami communicantes to the sympathetic trunk. A similar cell group is found in the second to fourth sacral segments but, unlike the thoracolumbar lateral cell column, it does not form a visible lateral projection. It is the source of the sacral outflow of parasympathetic preganglionic nerve fibres.

Ventral horn

Lamina IX neurones in the ventral horn vary in size. The largest cell bodies, which may exceed 25 μ m in diameter, are those of α motor neurones, the axons of which emerge in ventral roots to innervate extrafusal fibres in striated skeletal muscles. Large numbers of smaller neurones, 15–25 μ m in diameter, are also present. Some of these are γ motor neurones, which innervate intrafusal fibres of muscle spindles, and the rest are interneurons. Motor neurones utilize acetylcholine as their neurotransmitter.

The basic building block of the motor neurones in lamina IX is represented by a longitudinally disposed group of neurones, which innervate a given muscle, and in which the α and γ motor neurones are intermixed. The various groups innervating different muscles are aggre-

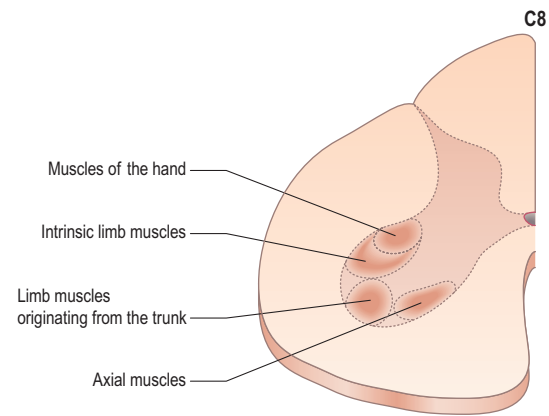


Fig. 20.7 The approximate location of motor neurone groups at the C8 segmental level of the spinal cord.

gated into two major longitudinal columns: medial and lateral. In transverse section these form the medial and lateral cell groups in the ventral horn (Fig. 20.7). The motor neurones of lamina IX are somatotopically organized. The basic arrangement is that medial cell groups innervate the axial musculature, and lateral cell groups innervate the limbs.

The medial nuclear group is divisible into a posteromedial and anteromedial group. This group innervates the muscles attached to the axial skeleton. The medial nuclear group extends throughout the length of the spinal cord. Its neurones innervate epaxial muscle groups (which extend the head and vertebral column) and hypaxial muscle groups (which flex the head and vertebral column). Epaxial muscles include the erector spinae group, while hypaxial muscles include prevertebral muscles of the neck, intercostal and anterior abdominal wall muscles. The epaxial muscles are innervated by branches of the dorsal primary rami of the spinal nerves, and the hypaxial muscles by branches of the ventral primary rami. In the medial column, motor neurones supplying epaxial muscles are sited ventral to those supplying hypaxial muscles.

The lateral nuclear group innervates the rest of the body musculature. In the thoracic region, it is small and innervates the lateral trunk muscles, including the intercostal muscles. In the cervical and lumbar enlargements, the lateral nuclear group is enlarged and a number of subgroups may be distinguished, including the anterolateral (ventrolateral), posterolateral (dorsolateral), central and retroposterolateral (retrodorsolateral) groups.

In the cervical and lumbar enlargements, the lateral nuclear group innervates muscles of the upper and lower limbs, respectively. These neurones are somatotopically organized. Passing round the cervical ventral horn clockwise from ventromedial to dorsolateral, α motor neurones successively innervate muscles of the trunk, shoulder girdle, upper arm and lower arm (see Fig. 20.4). The proximal muscles of the limb are supplied from motor cell groups located more rostrally in the enlargement than those supplying the distal muscles, e.g. motor neurones in C8 and T1 innervate the intrinsic muscles of the hand, and motor neurones in C5 and C6 innervate shoulder muscles. There is a similar overall arrangement in the lumbosacral cord for motor neurones innervating the muscles in the lower limb (Fig. 20.8).

The spinal nucleus of the accessory nerve (SAN) contains motor neurones that innervate sternocleidomastoid and trapezius. Described as a curvilinear band of motor neurones extending within the ventral horn from C1/2 to C5/6 (Orhan et al 2009; Saylam et al 2009) in line rostrally with the nucleus ambiguus, the spinal nucleus of the accessory nerve is unusual in that it not only lies between the most medial column in the ventral horn (innervating axial muscles) and the lateral column (innervating upper limb muscles) (Routal and Pal 2000) but also its rootlets exit the spinal cord just dorsal to the ligamentum denticulatum (Jenny et al 1988). They subsequently combine to form the intraspinal portion of the accessory nerve, which ascends the vertebral canal to enter the posterior cranial fossa via the foramen magnum. The supranuclear innervation of the spinal nucleus of the accessory nerve is uncertain. The bulk of current evidence favours bilateral hemispheric innervation of motor neurones in the rostral spinal nucleus of the accessory nerve (innervating sternocleidomastoid), although opinion varies as to the extent of the ipsilateral input, but predominantly contralateral supranuclear control of the neurones in the caudal spinal nucleus of the accessory nerve (innervating the upper and middle fibres of trapezius). This pattern is analogous to the supranuclear innervation of the facial nerve motor nucleus.

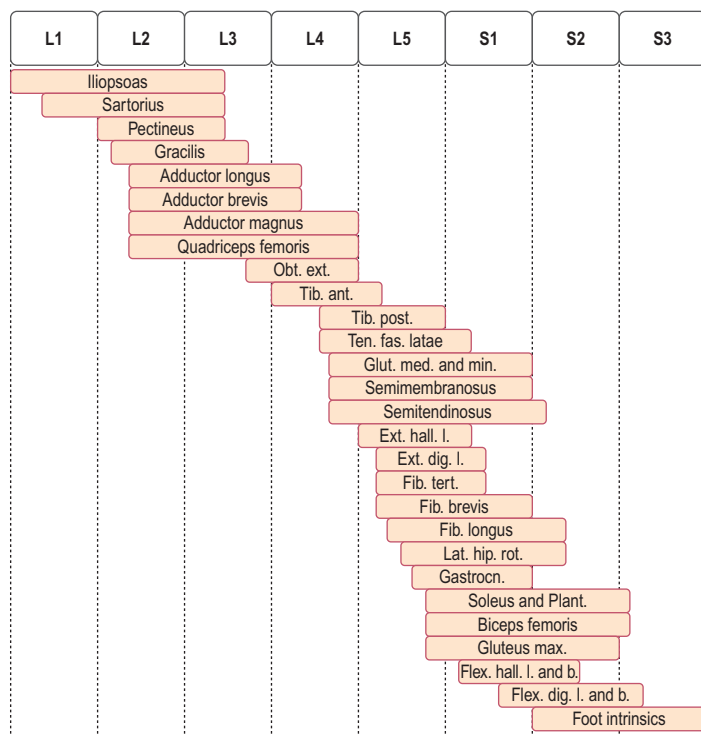


Fig. 20.8 The segmental arrangement of motor neurones innervating muscles of the lower limb. Obt. ext. = obturator externus; Tib. ant. = tibialis anterior; Tib. post. = tibialis posterior; Ten. fas. latae = tensor fasciae latae; Glut. med. and min. = gluteus medius and minimus; Ext. hall. l. = extensor hallucis longus; Fib. tert. = fibularis tertius; Fib. brevis = fibularis brevis; Fib. longus = fibularis longus; Lat. hip. rot. = lateral hip rotators; Gastrocn. = gastrocnemius; Soleus and plant. = soleus and plantaris; Gluteus max. = gluteus maximus; Flex. hall. l. and b. = flexors hallucis longus and brevis; Flex. dig. l. and b. = flexors digitorum longus and brevis.

The nucleus of Onuf, which is thought to innervate the perineal striated muscles, is an anterolateral group of lamina IX cells in the first and second sacral segments. The centrally situated phrenic nucleus, containing the motor neurones that innervate the diaphragm, lies in the third to seventh cervical segments. The lumbosacral nucleus is also centrally located and its axons have an unknown distribution.

The main afferent connections to motor neurones are: direct monosynaptic connections from proprioceptive dorsal root afferents in the same or nearby segments; connections from axonal collaterals of dorsal horn and other interneurones; direct monosynaptic connections from the vestibulospinal and corticospinal tracts.

SPINAL WHITE MATTER

The spinal white matter surrounds the central core of grey matter. The white matter consists primarily of longitudinally running nerve fibres. Fibres subserving related functions, or those with common origins or destinations, are generally grouped together anatomically to form tracts (fasciculi), which may be ascending, descending and propriospinal. Ascending tracts consist either of primary afferent fibres, which enter the cord via the dorsal roots of spinal nerves, or fibres derived from intrinsic spinal neurones, which carry afferent impulses to supraspinal levels. Descending tracts contain fibres that descend from the cerebral cortex or brainstem nuclei to control the activity of spinal neurones. Propriospinal tracts contain the axons of neurones that are localized entirely to the spinal cord; they contain both ascending and descending components, collectively mediating intersegmental coordination.

The spinal white matter is conventionally described as being arranged into three large, bilaterally paired masses, the dorsal, lateral and ventral funiculi, each of which contains a number of specific tracts (see Fig. 20.1; Fig. 20.9). The dorsal (posterior) funiculus lies between the dorsal horn and the posterior median septum. The lateral funiculus lies between the dorsal root entry zone and the site where the ventral roots emerge. The ventral (anterior) funiculus lies between the ventral median fissure and the site where the ventral roots emerge. Narrow dorsal and ventral white commissures run between the two halves of the cord.

Fibres in the spinal white matter vary in calibre. Many are small and lightly myelinated or unmyelinated. Most regions contain a wide spectrum of fibre diameters, from 1 μm or less to 10 μm . Some tracts typically contain only small fibres, e.g. the dorsolateral tract, fasciculus gracilis and central part of the lateral funiculus. The fasciculus cuneatus, ventral funiculus and peripheral zone of the lateral funiculus all contain many large-diameter fibres.

Whilst the ascending and descending tracts are, to a large extent, discrete and regularly located, significant overlap between adjacent tracts does occur. Their general disposition is shown in Figure 20.9 (Parent and Carpenter 1996).

Ascending pathways

Dorsal columns

The dorsal funiculus consists of two large ascending tracts, the fasciculus gracilis and fasciculus cuneatus (Fig. 20.10), which are also known as the dorsal columns. They are separated by a posterointermediate septum. The dorsal columns contain a high proportion of myelinated fibres carrying proprioceptive (position sense and kinaesthesia), exteroceptive (touch-pressure) and vibratory sensation to higher levels. These fibres come from several sources: long primary afferent fibres, which enter the cord in the dorsal roots of spinal nerves and ascend to the dorsal column nuclei in the medulla oblongata; shorter primary afferent fibres projecting to neurones of the posterior thoracic nucleus and other spinal neurones; and axons from secondary neurones of the spinal cord ascending to the dorsal column nuclei. The dorsal columns also contain axons of propriospinal neurones.

The fasciculus gracilis begins at the caudal end of the spinal cord. It contains long ascending branches of primary afferents, which enter the cord through ipsilateral dorsal spinal roots and ascending axons of secondary neurones in laminae IV–VI of the ipsilateral dorsal horn. As the fibres ascend, they are joined by axons of successive dorsal roots. Fibres entering in coccygeal and lower sacral regions are shifted medially by successive additions of fibres entering at higher levels.

The fasciculus gracilis lies medial to the fasciculus cuneatus in the upper spinal cord (see Fig. 20.9). At upper cervical levels, the fasciculus gracilis contains a larger proportion of afferents from cutaneous receptors than from deep proprioceptors because many of the latter leave the fasciculus at lower segments to synapse in the posterior thoracic nucleus. Indeed, proprioception from the lower limb mostly reaches the thalamus by relaying in the posterior thoracic nucleus and then again in nucleus Z. Axons of the fasciculus gracilis, from both primary and secondary neurones, terminate in the nucleus gracilis of the dorsal medulla.

The fasciculus cuneatus (see Fig. 20.9) begins at mid-thoracic level and lies lateral to the fasciculus gracilis. It is composed mostly of primary afferent fibres of the upper thoracic and cervical dorsal roots. At upper cervical levels, it contains a large population of afferents from both deep and cutaneous receptors of the upper limb. In addition, some of its axons arise from secondary neurones in laminae IV–VI of the ipsilateral dorsal horn. Many axons (both primary and secondary) that ascend in the fasciculus cuneatus terminate in the nucleus cuneatus of the dorsal medulla. Some also end in the lateral (external or accessory) cuneate nucleus; neurones in this nucleus project to the cerebellum via the cuneocerebellar pathway.

Many ascending fibres of the fasciculus gracilis and fasciculus cuneatus terminate by synapsing on neurones of the dorsal column nuclei (nucleus gracilis and nucleus cuneatus, respectively) in the medulla oblongata. (The connections of the dorsal column nuclei are described further with the medulla oblongata; Ch. 21.) Axons arising from neurones in the dorsal column nuclei arch ventromedially round the central grey matter of the medulla as internal arcuate fibres (see Fig. 21.7) and decussate in the great sensory decussation to form the medial lemniscus. They ascend to the ventral posterolateral nucleus of the thalamus, from where neurones project to the somatosensory cortex in the postcentral gyrus of the parietal lobe (areas 1, 2 and 3). Some neurones of the dorsal column nuclei form posterior external arcuate fibres that enter the cerebellum.

The high degree of somatotopic organization that is present in the dorsal columns is preserved as the pathways ascend through the dorsal column nuclei and thalamus to reach the primary somatosensory cortex. In the dorsal column nuclei, the lower limb is represented in the nucleus gracilis, the upper limb in the nucleus cuneatus, and the trunk in an intermediate position between them. Fibres are also segregated by modality in the dorsal columns; fibres from hair receptors are most superficial, while those from tactile and vibratory receptors lie in deeper layers.

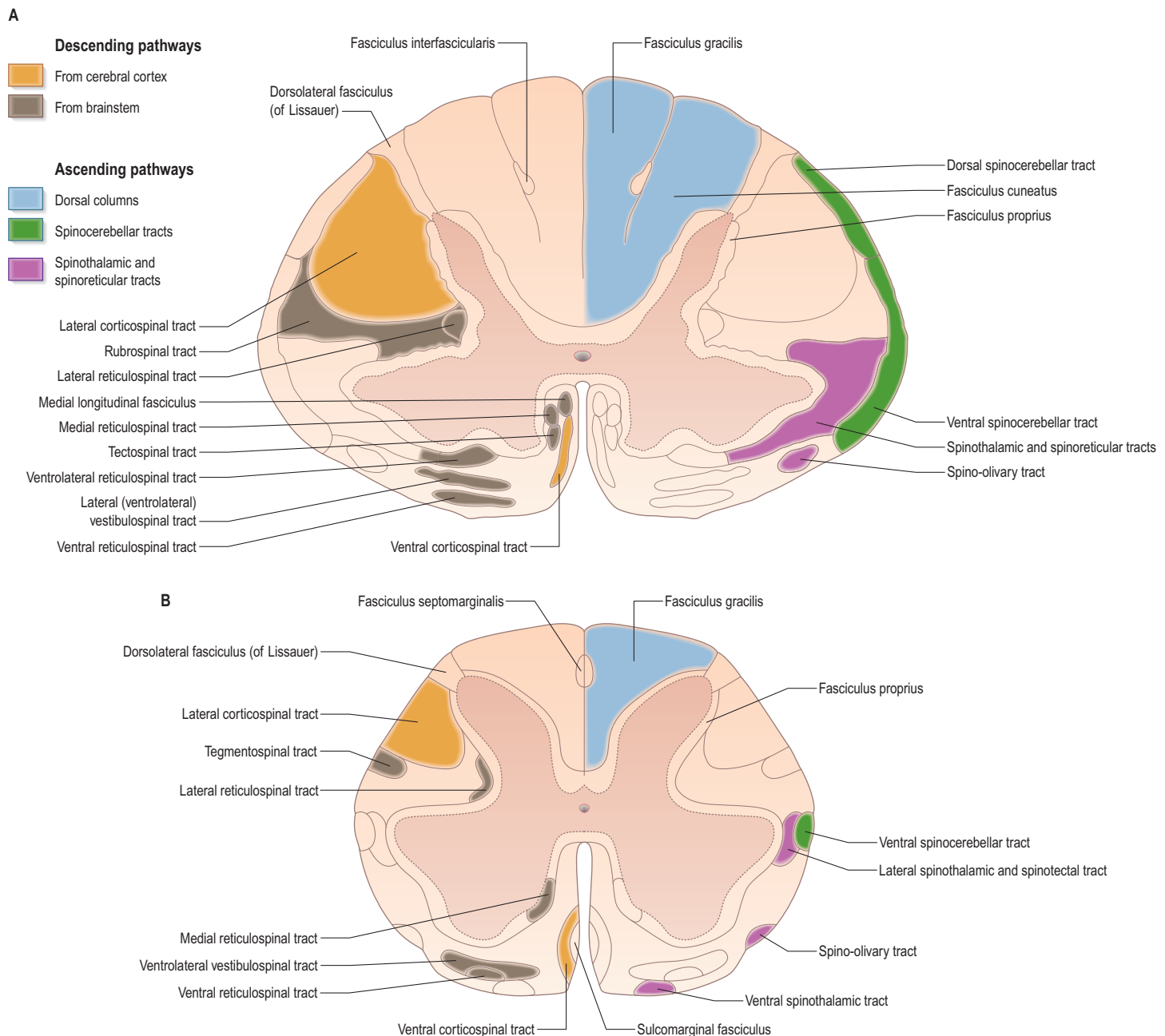


Fig. 20.9 The approximate positions of nerve fibre tracts in the spinal cord at mid-cervical (**A**) and lumbar (**B**) levels.

Spinocerebellar tracts

The dorsal (posterior) and ventral (anterior) spinocerebellar tracts occupy the periphery of the lateral aspect of the spinal white matter (see Fig. 20.9; Fig. 20.11) and carry proprioceptive and cutaneous information to the cerebellum for the coordination of movement. Both tracts contain large-diameter myelinated fibres but there are more in the dorsal tract. Finer-calibre fibres are associated with the ventral tract.

The dorsal spinocerebellar tract lies lateral to the lateral corticospinal tract. It begins at about the level of the second or third lumbar segment and enlarges as it ascends. Axons of the tract originate ipsilaterally from the larger neurones of the posterior thoracic nucleus, in lamina VII throughout spinal segments T1–L2. The posterior thoracic nucleus receives input from collaterals of long ascending primary afferents of the dorsal columns and terminals of shorter ascending primary afferents of the dorsal columns. Many of these afferent fibres ascend from segments caudal to L2. In the medulla, the dorsal spinocerebellar tract passes through the restiform body and thence the inferior cerebellar peduncle to terminate ipsilaterally in the rostral and caudal parts of the cerebellar vermis.

The ventral spinocerebellar tract lies immediately ventral to the dorsal tract. The cells of origin are in laminae V–VII of the lumbosacral cord and the tract carries information from the lower limb. Most of the axons forming the tract decussate but some remain ipsilateral. The tract begins in the upper lumbar region and ascends through the medulla

oblongata to reach the upper pontine level, from where it descends in the dorsal part of the superior cerebellar peduncle to terminate, mainly contralaterally, in the anterior cerebellar vermis.

The rostral spinocerebellar tract originates from cell groups of the intermediate zone and horn of the contralateral cervical enlargement and is considered to be the upper limb and upper trunk equivalent of the ventral spinocerebellar tract.

Experimental tracing studies in animals have demonstrated a cervico-spinocerebellar tract arising from a central cervical nucleus in the upper cervical spinal cord (C1–C4); the central cervical nucleus, possibly Cajal's nucleus intermedius, receives primary afferents from the vestibular labyrinth and muscle afferents from the deep muscles of the neck. A similar pathway may exist in humans where the nucleus intermedius of Cajal receives fibres from only the upper two cervical roots (Marani and Schoen 2005).

The dorsal and ventral spinocerebellar tracts are organized such that fibres from lower spinal segments are most superficial. Both tracts convey proprioceptive and exteroceptive information, but they are functionally different. Neurones of the posterior thoracic nucleus are excited monosynaptically by Ia and Ib primary afferent fibres (from muscle spindles and tendon organs, respectively) and also by group II muscle afferents, and cutaneous touch and pressure afferents. The proprioceptive impulses often arise from a single muscle or from synergistic muscles acting at a common joint. Thus, the dorsal spinocerebellar tract transmits modality-specific and space-specific information that is used

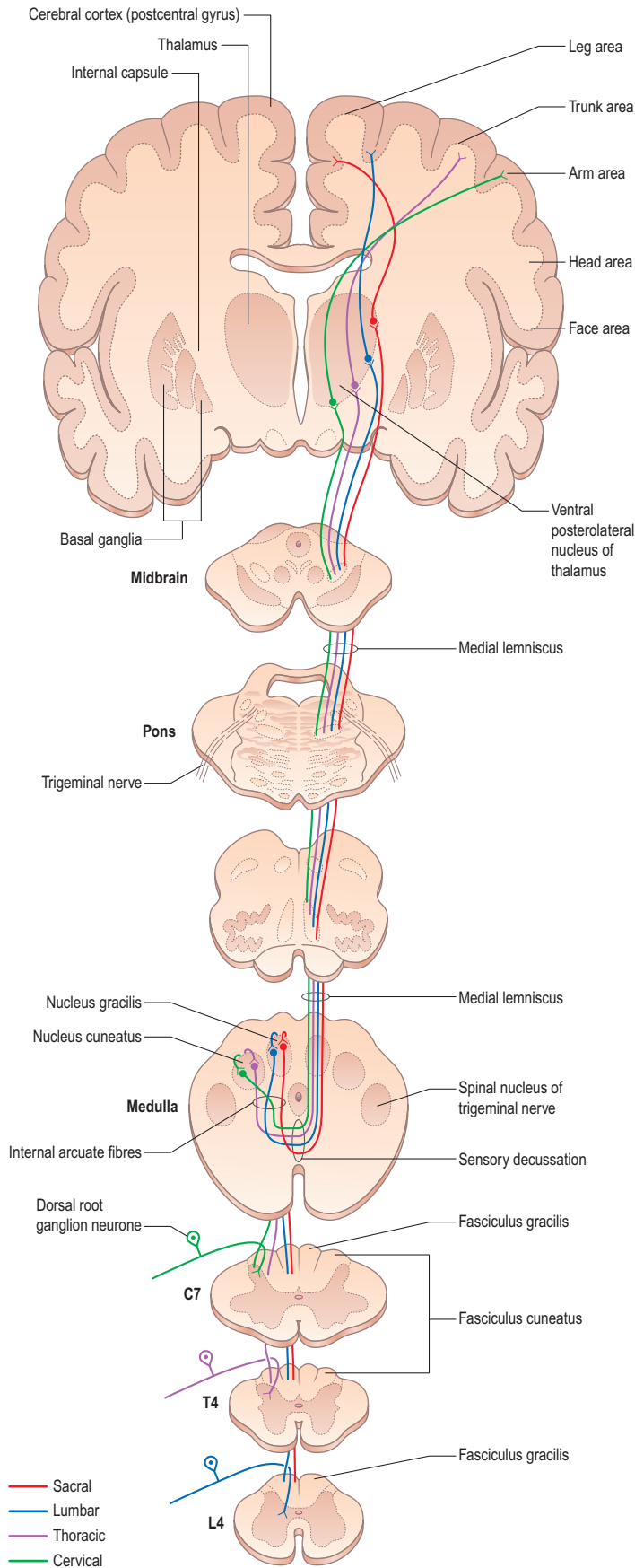


Fig. 20.10 The dorsal columns. Primary afferent fibres from different levels and their associated second- and third-order neurones are depicted in different colours.

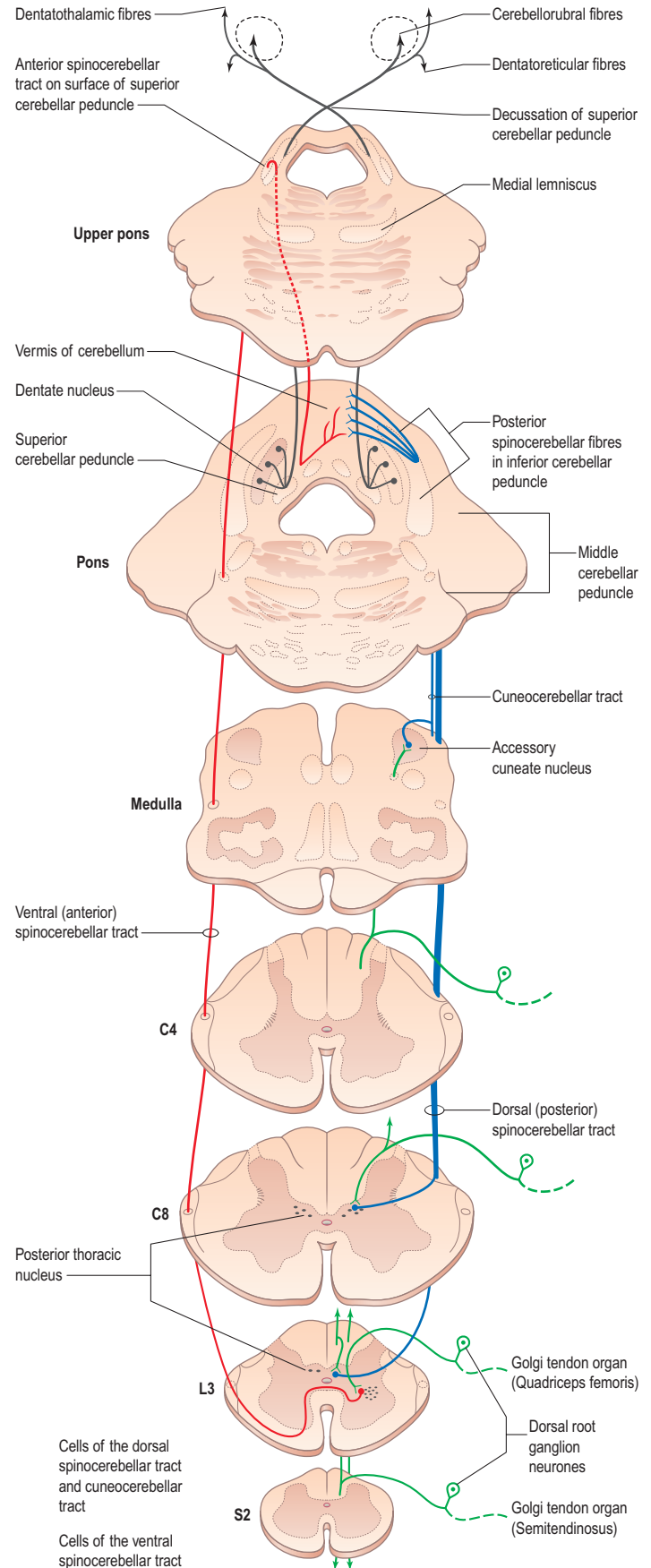


Fig. 20.11 The spinocerebellar tracts. (Redrawn with permission from Carpenter MB 1991 Core Text of Neuroanatomy, 4th edn. Baltimore: Williams and Wilkins.)

in the fine coordination of individual limb muscles. On the other hand, the cells of the ventral tract are activated monosynaptically by Ib afferents and transmit information from large receptive fields that include different segments of a limb. The ventral tract lacks subdivisions for different modalities and transmits information for the coordinated movement and posture of the entire lower limb.

Since the posterior thoracic nucleus diminishes rostrally (see Fig. 20.4) and does not extend above the lowest cervical segment, it follows that the dorsal spinocerebellar tract carries information from the trunk and lower limb. Proprioceptive and exteroceptive information from the upper limb travel in primary afferent fibres of the fasciculus cuneatus. These fibres end somatotopically in the accessory (external or lateral)

cuneate nucleus and the adjoining part of the cuneate nucleus situated in the medulla oblongata. Cells of these nuclei give rise to the posterior external arcuate fibres that form the cuneocerebellar tract (see Fig. 20.11), which enters the cerebellum via the ipsilateral inferior cerebellar peduncle. The cells in the accessory cuneate nucleus and the lateral part of the cuneate nucleus are considered to be homologous to the cells in the posterior thoracic nucleus. The cuneocerebellar tract is, therefore, functionally allied to the dorsal spinocerebellar tract, and is its upper-limb equivalent.

Axons of all the spinocerebellar tracts and the cuneocerebellar tract form part of the 'mossy-fibre system'. They end in the cerebellar cortex in a highly organized, somatotopic and functional pattern.

Spinothalamic tracts

The spinothalamic tracts (see Fig. 20.9) consist of second-order neurones that convey pain, temperature, coarse (non-discriminative) touch and pressure information to the somatosensory region of the thalamus. The cells of origin lie in various laminae of all segments of the spinal cord. Fibres decussate in the ventral white commissure to reach the contralateral spinothalamic tracts; pain and temperature fibres do so promptly, within about one segment of their origin, whilst fibres carrying other modalities may ascend for several segments before crossing. Spinothalamic fibres mostly ascend in the white matter ventrolateral to the ventral horn, partly intermingled with ascending spinoreticular fibres and descending reticulospinal fibres. Some authorities describe two spinothalamic tracts (lateral and ventral) with more or less distinct anatomical locations and functions. However, physiological studies in animals support the notion that these tracts may be best considered as a structural and functional continuum.

The lateral spinothalamic tract (Fig. 20.12) is sited in the lateral funiculus, lying medial to the ventral spinocerebellar tract. Clinical evidence indicates that it subserves pain and temperature sensations. The ventral spinothalamic tract (Fig. 20.13) lies in the ventral funiculus medial to the point of exit of the ventral spinal nerve roots and dorsal to the vestibulospinal tract, which it overlaps. On the basis of clinical evidence, it subserves coarse tactile and pressure modalities.

A dorsolateral spinothalamic tract has been described in animals. The axons arise mainly from neurones in lamina I, and cross to ascend in the contralateral dorsolateral funiculus. These neurones respond maximally to noxious, mechanical and thermal cutaneous stimuli. Examples of clinical pain relief following dorsolateral cordotomy suggest that a similar projection exists in humans.

On reaching the lower brainstem, spinothalamic tract axons separate. Axons in the ventral tract join the medial lemniscus. Axons in the lateral tract continue as the spinal lemniscus.

There is clear somatotopic organization of the fibres in the spinothalamic tracts throughout their extent. Fibres crossing at any cord level join the deep aspect of those that have already crossed, which means that both tracts are segmentally laminated (Fig. 20.14). Somatotopy is maintained throughout the medulla oblongata and pons. In the mid-brain, fibres in the spinal lemniscus conveying pain and temperature sensation from the lower limb extend dorsally, while those from the trunk and upper limb are more ventrally placed. Both lemnisci ascend to end in the thalamus. The major spinothalamic projections in humans are to the ventral posterolateral nucleus, and also to the centrolateral intralaminar nucleus.

Neurones of the spinothalamic tracts

The specific localization of spinothalamic tract cell bodies is poorly documented in humans. In animals, about one-third are localized to the upper three cervical segments. About 20% are located in lower cervical segments, 20% in the thoracic region (mostly in segments T1–3), 20% in the lumbar region and 10% in the sacrococcygeal cord. Cells are located in laminae I and IV–VIII, the greatest concentration being in laminae VI and VII. Cell bodies giving rise to spinothalamic tract axons are predominantly contralateral, although a relatively small number (10%), mostly in the upper three cervical segments, are ipsilateral.

Neurones of the spinothalamic tracts have varied receptive fields. Specificity of separate types of afferents, as it exists in the dorsal column nuclei, is absent in the spinothalamic nuclei. Convergence of different functional types of afferent fibres on to an individual tract cell is a common feature in the cord. On the basis of laminar site, functional properties, and specific thalamic termination of their axons, spinothalamic tract neurones may be divided into three separate groups. These are the apical cells of the dorsal grey column (lamina I), deep dorsal column cells (laminae IV–VI), and cells in the ventral grey column (laminae VII and VIII). There are species differences and the description below is derived from studies in non-human primates.

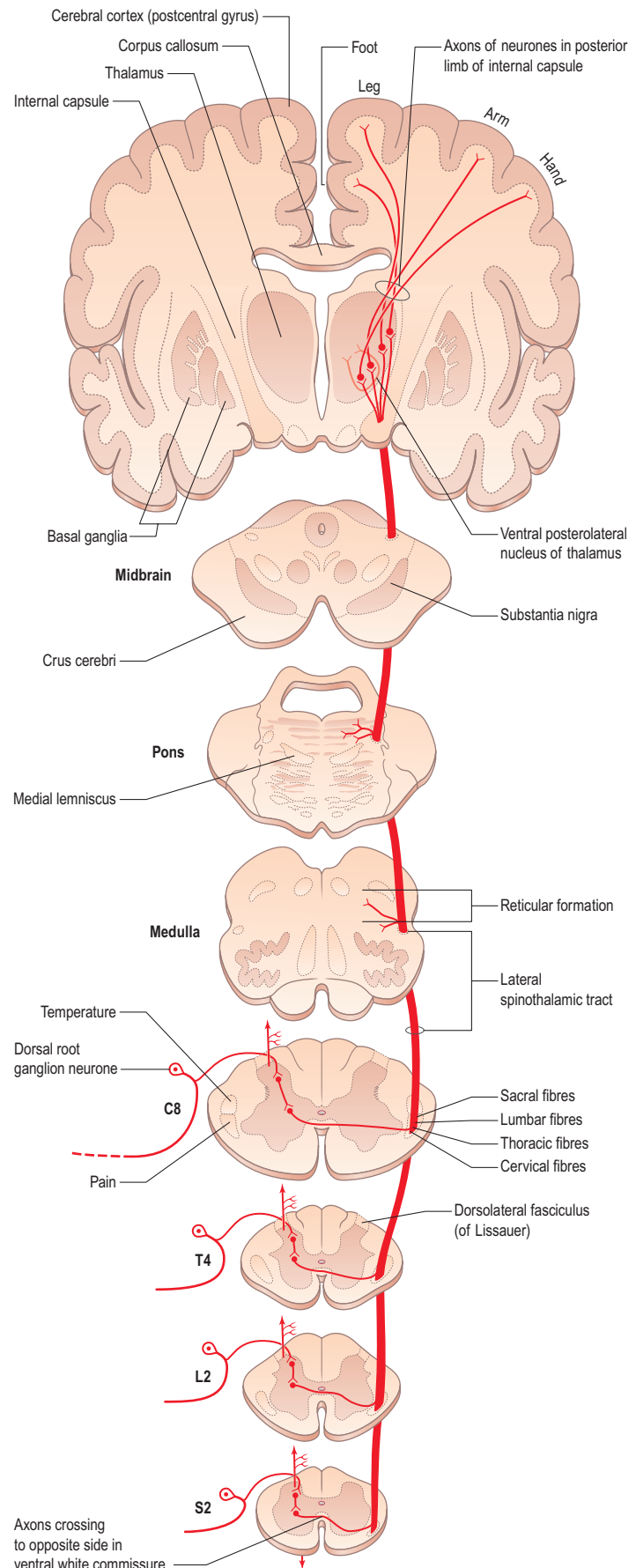


Fig. 20.12 The lateral spinothalamic tract. (Redrawn with permission from Carpenter MB 1991 *Core Text of Neuroanatomy*, 4th edn. Baltimore: Williams and Wilkins.)

Lamina I cells project to the thalamus. They respond maximally to noxious or thermal cutaneous stimulation and consist mainly of high-threshold but also some wide-dynamic-range units (responding to various different intensities of stimulus). Their receptive fields are usually small, representing a part of a digit or a small area of skin involving several digits. Lamina I spinothalamic tract neurones receive

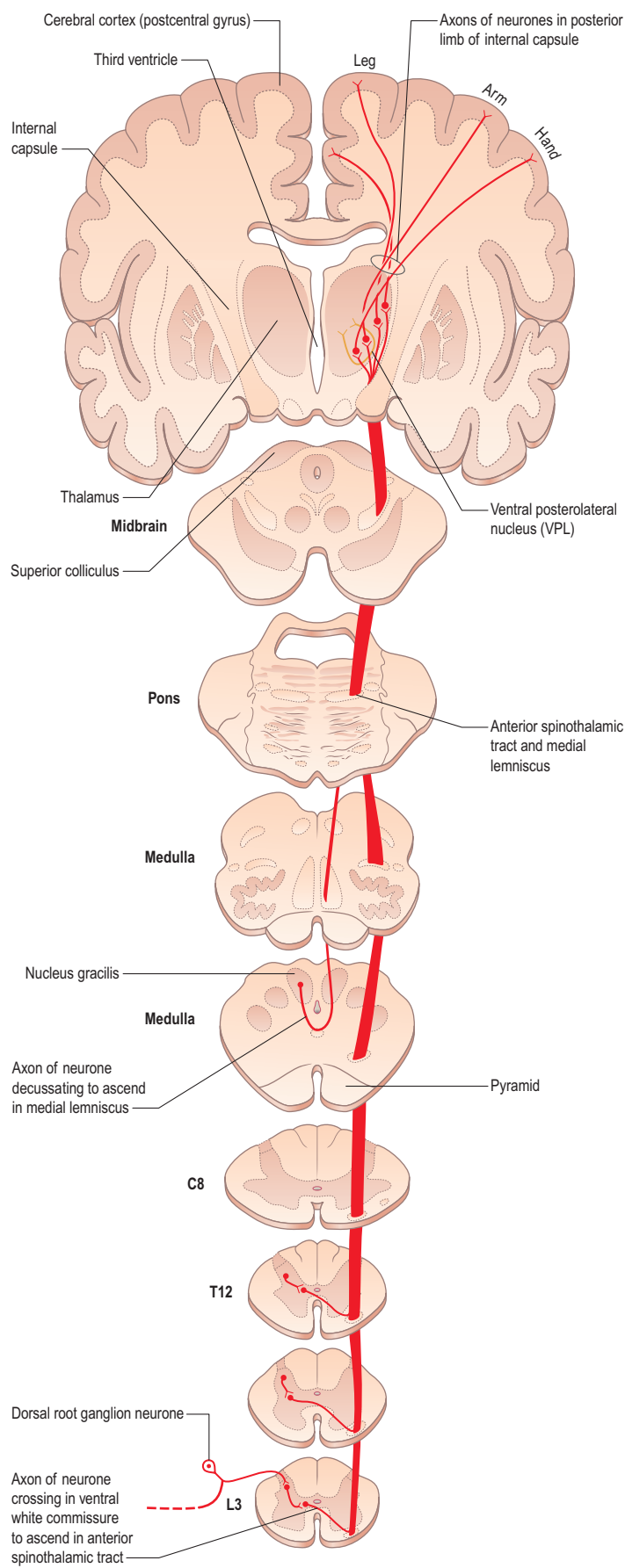


Fig. 20.13 The ventral (anterior) spinothalamic tract. (Redrawn with permission from Carpenter MB 1991 *Core Text of Neuroanatomy*, 4th edn. Baltimore: Williams and Wilkins.)

input from A δ and C fibres, and some respond to convergent input from deep somatic and visceral receptors. Spinothalamic tract cells in the thoracic cord display marked viscerosomatic convergence. Hence pain from cardiac disease such as angina or a myocardial infarction is commonly felt as a tightness around the chest wall (referred pain). This is very similar to the pain that may be felt after an infarct of the spinal

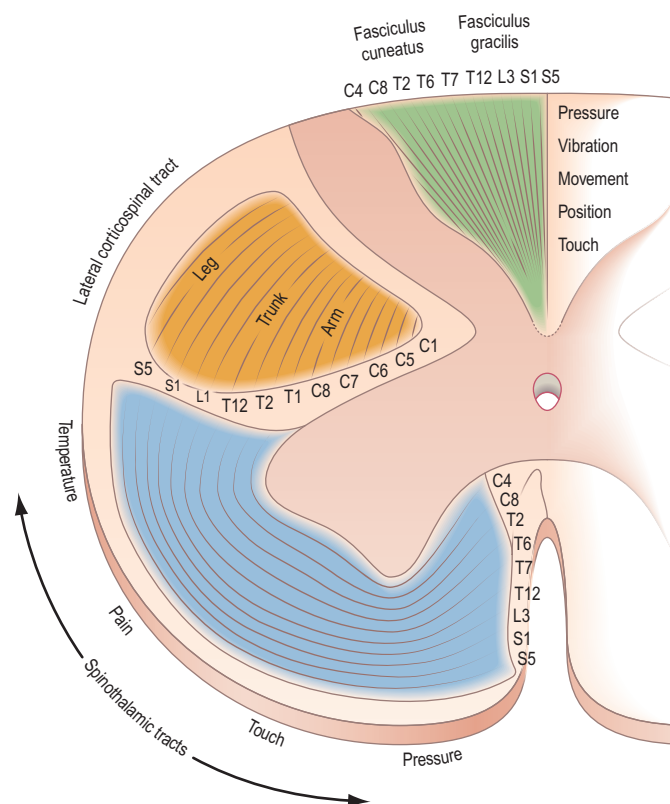


Fig. 20.14 The general plan of the segmental organization of fibres in the dorsal funiculus, the lateral corticospinal tract and the spinothalamic tracts. The probable cross-sectional areas of these tracts are schematically enlarged. This general plan applies to all segmentally organized tracts, whether ascending, descending, ipsilateral or contralateral.

cord (Combarros et al 2002). Lamina I spinothalamic tract neurones project preferentially to the ventral posterolateral nucleus of the thalamus, with limited projections to the centrolateral intralaminar and mediodorsal thalamic nuclei.

The population of deep dorsal column (laminae IV–VI) spinothalamic neurones of the lumbar cord contains units of the wide-dynamic-range (60%), high-threshold (30%) and low-threshold (10%) types. They can accurately code both innocuous and noxious cutaneous stimuli. Some cells also respond to input from deep somatic and visceral receptors. In the lumbar cord, their receptive fields are small or medium-sized; they are larger than the area of the foot but smaller than the entire leg. In the thoracic cord, the fields of these laminar cells are larger and often include the entire upper limb plus part of the chest. Many of the deep dorsal grey column spinothalamic tract neurones in the thoracic segments receive convergent input from sympathetic afferent fibres.

Laminae IV–VI spinothalamic tract units project either to the ventral posterolateral nucleus or to the centrolateral intralaminar nucleus of the thalamus, and sometimes to both. Units projecting to the ventral posterolateral nucleus receive input from all classes (A β , A δ and C) of cutaneous fibres.

Ventral grey column (laminae VII and VIII) spinothalamic tract cells respond mainly to deep somatic (muscle and joint) stimuli but also to innocuous and/or noxious cutaneous stimuli. In the thoracic regions of the spinal cord, they also receive convergent input from visceral sources. The majority of laminae VII and VIII spinothalamic tract neurones have large, complex receptive fields (often bilateral), which encompass widespread areas of the body. Cells of this group, which project exclusively to the medial thalamus, receive input from A β , A δ and C classes of afferent fibres, and many respond to convergent input from receptors of deep structures. This population of neurones contains wide-dynamic-range (25%), high-threshold (63%) and low-threshold (12%) units. Most of the spinothalamic tract cells in the ventral grey column project to the intralaminar nuclei of the thalamus. The wide-dynamic-range type of neurones are particularly effective for discriminating between different intensities of painful stimulation.

Spinoreticular pathway

Spinoreticular fibres are intermingled with those of the spinothalamic tracts and ascend in the ventrolateral quadrant of the spinal cord

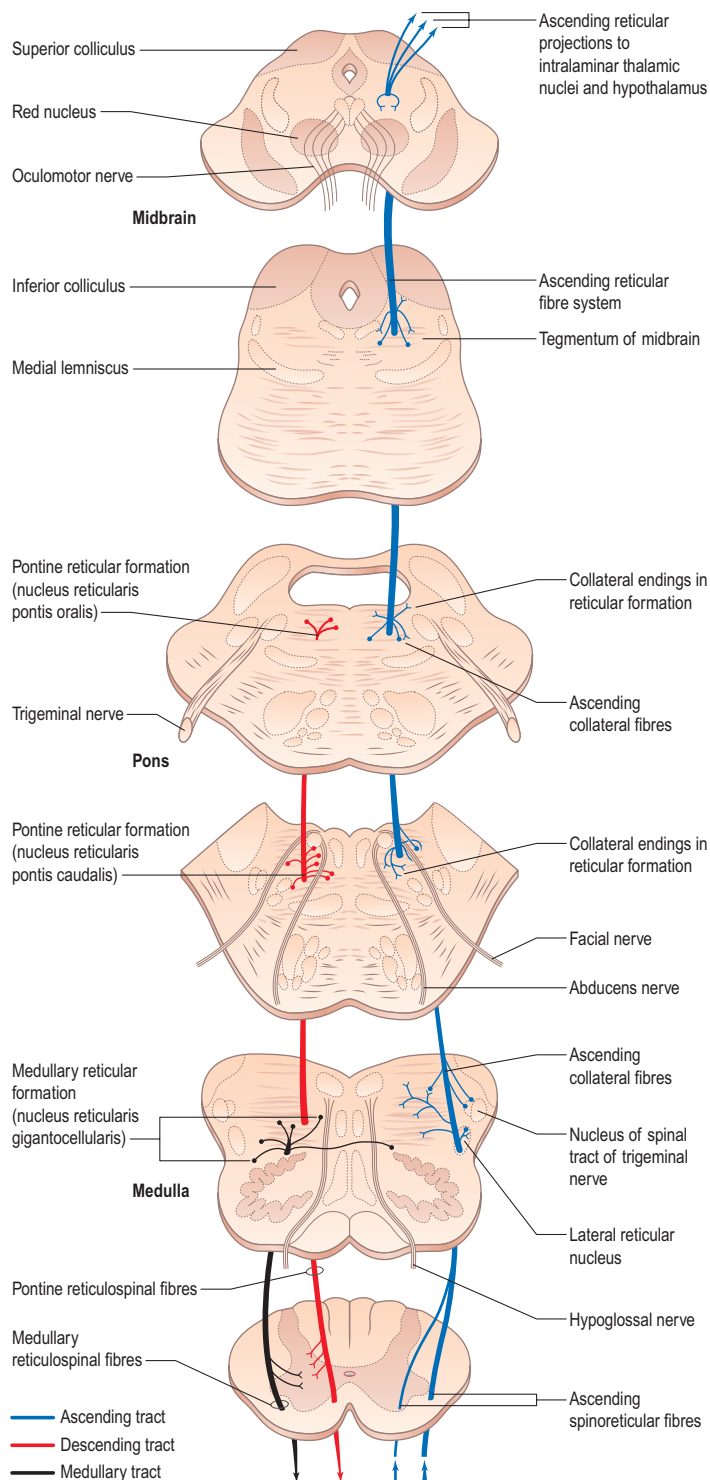


Fig. 20.15 The reticular tracts. (Redrawn with permission from Carpenter MB 1991 *Core Text of Neuroanatomy*, 4th edn. Baltimore: Williams and Wilkins.)

(Fig. 20.15). Evidence from animal studies suggests that cells of origin occur at all levels of the spinal cord, particularly in the upper cervical segments. Most neurones are in lamina VII, some are in lamina VIII, and others are in the dorsal horn, especially lamina V. Most axons in the lumbar and cervical enlargements cross the midline, but there is a large uncrossed component in cervical regions. Most axons are myelinated. The pattern of anterograde degeneration, in both human post-mortem studies and in experimental animals following anterolateral cordotomy, indicates the existence of spinoreticular projections to many nuclei of the medial pontomedullary reticular formation. There is also a projection to the lateral reticular nucleus (a precerebellar relay nucleus). These projections do not appear to be somatotopically organized. Spinoreticular neurones respond to inputs from the skin or deep tissues. Innocuous cutaneous stimuli may inhibit or excite a particular cell, whereas noxious stimuli are often excitatory. A spino-reticulo-thalamo-cortical pathway has been proposed as an important route

serving pain perception. Like other ascending pathways, the tract cells are influenced by descending control. For example, electrical stimulation of the periaqueductal grey matter inhibits the responses of certain spinoreticular cells to input from cardiopulmonary afferents. Stimulation of the reticular formation also alters the activity of spinoreticular neurones.

Spinocervicothalamic pathway

The lateral cervical nucleus is small in humans. It lies in the lateral funiculus, ventrolateral to the dorsal horn in the upper two cervical segments. In some human cord specimens, the nucleus is not distinctly defined and is possibly incorporated into the dorsal horn. It receives axons from the spinocervical tract, which ascends in the dorsolateral funiculus. The tract cells are found in laminae III–V at all levels of the spinal cord, ipsilateral to the nucleus. Most neurones of the nucleus project to the contralateral thalamus via the medial lemniscus, and some project to the contralateral midbrain. Specific thalamic targets include the ventral posterolateral nucleus and part of the posterior complex. Spinocervical tract neurones respond to hair movement, pressure, pinch and thermal stimuli and to high-threshold muscle input; many also respond to noxious stimuli. Like tract cells of other ascending pathways, they are under tonic descending inhibitory control.

Spinomesencephalic pathway

The spinomesencephalic pathway consists of a number of tracts ascending from the spinal cord to various regions of the midbrain. It includes the spinotectal tract projecting to the superior colliculus, neurones synapsing in the periaqueductal grey matter, and other spinal cord projections that terminate in the parabrachial nucleus, the pretectal nuclei and the nucleus of Darkschewitsch. Cells of origin are located throughout the length of the spinal cord, particularly in the cervical segments and the lumbosacral enlargement, mostly in lamina I, but they are also present in laminae IV–VIII, where they are concentrated in lamina V. Most are contralateral, but a prominent ipsilateral group is also found at upper cervical levels. Spinomesencephalic fibres are mostly myelinated and ascend in the white matter of the ventrolateral quadrant of the spinal cord, in association with the spinothalamic and spinoreticular tracts.

Spinomesencephalic neurones are of low-threshold, wide-dynamic-range or high-threshold classes. Their receptive fields may be small, or very complex and encompass large surface areas of the body. Many spinomesencephalic cells are nociceptive and are likely to be involved in the motivational-affective component of pain. Electrical stimulation of their site of termination in the periaqueductal grey matter results in severe pain in humans. Furthermore, the cells of the deeper layers of the superior colliculus, where spinotectal fibres synapse, are activated by noxious stimuli.

Spino-olivary tract

The spino-olivary tract is described in animals as arising from neurones in the deeper laminae of grey matter. Axons forming the tract cross and then ascend superficially at the junction of the ventral and lateral white funiculi, to end in the dorsal and medial accessory olivary nuclei. The tract carries information from muscle and tendon proprioceptors, and also from cutaneous receptors. A functionally similar route, the dorsal spino-olivary tract, ascends in the dorsal white funiculi, and relays in the dorsal column nuclei to the contralateral inferior olivary nucleus. Information on these tracts in primates is scant, but postmortem evidence following cordotomies in humans has revealed degenerating axonal terminals in the inferior olivary nucleus.

Descending tracts

Descending pathways to the spinal cord originate primarily from the cerebral cortex and from numerous sites within the brainstem (see Fig. 20.15; Figs 20.16–20.17). They are concerned with the control of movement, muscle tone and posture, the modulation of spinal reflex mechanisms and the modulation of transmission of afferent information to higher levels. They also mediate control over spinal autonomic neurones (Kuypers 1981).

Corticospinal and corticonuclear tracts

Corticospinal and corticonuclear fibres arise from neurones in the cerebral cortex. They project, in a somatotopically organized fashion, to neurones that are mostly located in the contralateral spinal cord or brainstem respectively (see Fig. 20.16). The majority of corticospinal and corticonuclear fibres arise from cells situated in the primary motor cortex (area 4) and the premotor cortex (area 6). A small contribution

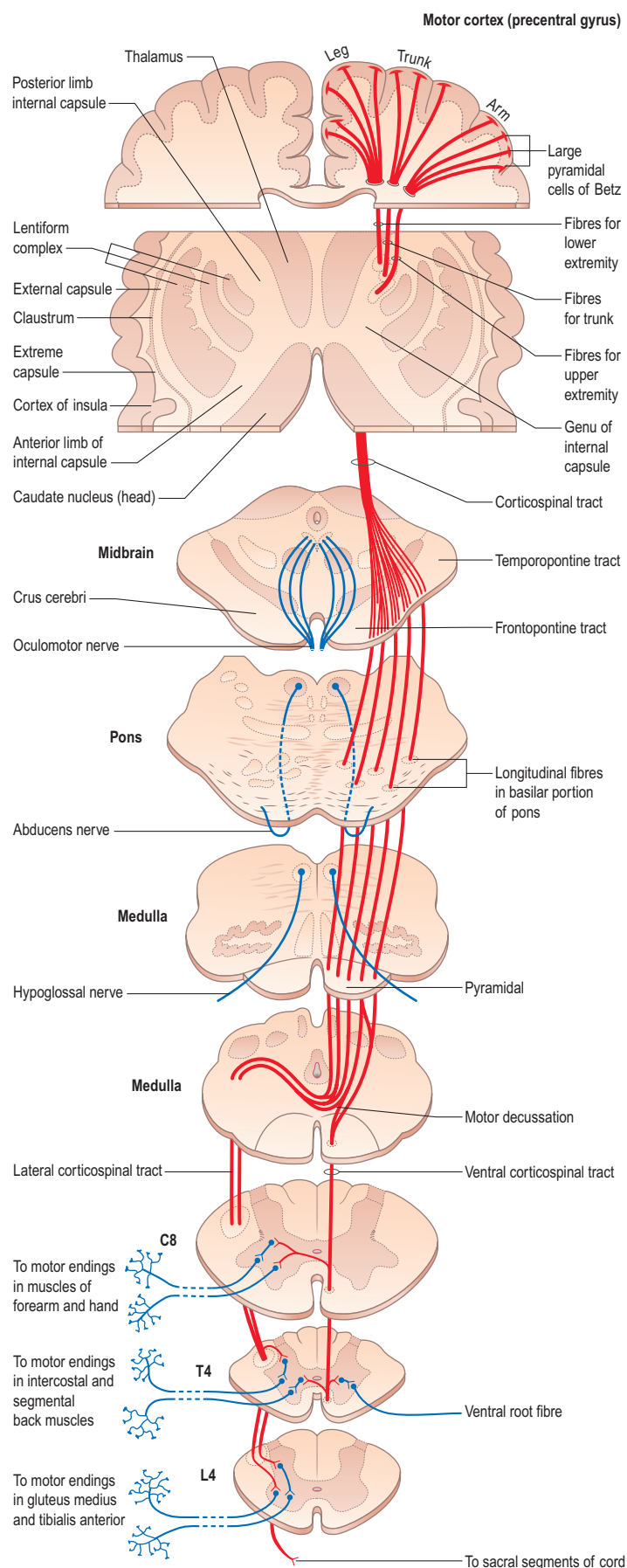


Fig. 20.16 The corticospinal tracts. (Redrawn with permission from Carpenter MB 1991 Core Text of Neuroanatomy, 4th edn. Baltimore: Williams and Wilkins.)

comes from cells in the postcentral gyrus (somatosensory cortex; areas 3, 1 and 2) and the adjacent parietal cortex (area 5). In the monkey, 30% of corticospinal fibres arise from area 4, 30% from area 6, and 40% from the parietal regions. The cells of origin of corticospinal and corticonuclear fibres vary in size according to their cortical origins and

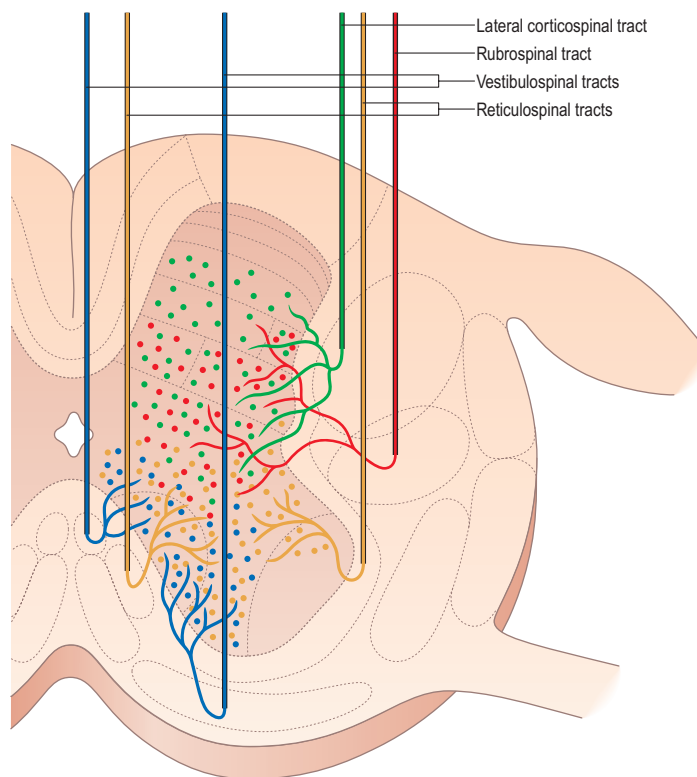


Fig. 20.17 A simplified scheme of some of the major descending pathways of the spinal cord, including their overlapping zones of termination in the grey matter. Within the grey matter the dotted lines show the laminar pattern, while within the white matter they are an approximate guide to the topography of the tracts.

are clustered into groups or strips. The largest neurones (giant pyramidal neurones, Betz cells) are located in the primary motor cortex of the precentral gyrus.

Corticospinal and corticonuclear fibres descend through the subcortical white matter to enter the genu and posterior limb of the internal capsule. They then pass through the ventral part of the midbrain in the crus cerebri. As they continue caudally through the pons, they are separated from its ventral surface and fragmented into fascicles by transversely running pontocerebellar fibres. Corticonuclear fibres leave to terminate in association with the cranial nerve motor nuclei of the midbrain, pons and medulla. In the medulla oblongata, the residual corticospinal fibres form a discrete bundle, the pyramid, which is visible as a prominent longitudinal column on the ventral surface of the medulla (see Fig. 21.4); the corticospinal tract is, therefore, also referred to as the pyramidal tract. Each pyramid contains about a million axons of varying diameter. The majority are myelinated. Most have a diameter of 1–4 μm ; approximately 10% have diameters of 5–10 μm ; and a very few have diameters of 11–22 μm . The largest-diameter axons arise from the giant pyramidal neurones.

Just rostral to the level of the spinomedullary junction, 75–90% of the corticospinal fibres in the pyramid cross the median plane in the motor decussation (decussation of the pyramids) and continue caudally as the lateral corticospinal tract. The rest of the fibres continue uncrossed as the ventral corticospinal tract. The lateral corticospinal tract (see Fig. 20.16) descends in the lateral funiculus throughout most of the length of the spinal cord until about the fourth sacral segment. It occupies an oval area, ventrolateral to the dorsal horn and medial to the dorsal spinocerebellar tract (see Fig. 20.9). In the lumbar and sacral regions, where the dorsal spinocerebellar tract is absent, the lateral corticospinal tract reaches the dorsolateral surface of the cord. The lateral corticospinal tract is somatotopically organized such that axons destined for spinal segments innervating the upper limbs are located more medially than those destined for cord segments innervating the lower limbs (see Fig. 20.14). As it descends, its fibres terminate in progressively lower spinal segments, which means that the tract diminishes in size. The lateral corticospinal tract also contains some uncrossed corticospinal fibres.

The smaller ventral corticospinal tract (see Fig. 20.16) descends in the ventral funiculus. It lies close to the ventral median fissure, and is separated from it by the sulcomarginal fasciculus (see Fig. 20.9). The ventral corticospinal tract diminishes as it descends and usually disappears completely at mid-thoracic cord levels. It may either be absent or,

very rarely, contain almost all the corticospinal fibres. Near their termination, most fibres of the tract cross the median plane in the ventral white commissure to synapse on contralateral neurones. The vast majority of corticospinal fibres, irrespective of the tract in which they descend, therefore terminate in the spinal cord on the side contralateral to their cortical origin.

Knowledge of the detailed termination of corticospinal fibres is based largely on animal studies, but is supplemented by data from postmortem studies on human brains using anterograde degeneration methods. Most corticospinal fibres are believed to terminate contralaterally on interneurons in the lateral parts of laminae IV–VI and both lateral and medial parts of lamina VII; some are also distributed to lamina VIII bilaterally. Terminals are also associated with contralateral motor neuronal cell groups in lamina IX, in the dorsolateral group and the lateral parts of both central and ventrolateral groups (see Fig. 20.17).

Corticospinal fibres from the frontal cortex, including motor and premotor areas 4 and 6, terminate mostly on interneurons in laminae V–VIII, with the densest concentration ending in the lateral part of lamina VI. They influence α and γ motor neurones of lamina IX, via these interneurons. As the widespread dendrites of multipolar neurones in lamina IX penetrate lamina VII, direct monosynaptic axodendritic contacts also occur on large α motor neurones. Direct termination on motor neurones is most abundant in the spinal enlargements.

Experimental evidence shows that precentral corticospinal axons influence the activities of both α and γ motor neurones, facilitating flexor muscles and inhibiting extensors, which are the opposite effects to those mediated by lateral vestibulospinal fibres. Evidence from animal studies shows that direct projections from the precentral cortical areas to spinal motor neurones are concerned with highly fractionated, precision movements of the limbs. Accordingly, in primates, precentral corticospinal fibres are mainly distributed to motor neurones supplying the distal limb muscles. Corticospinal projections may use glutamate or aspartate, often co-localized, as excitatory neurotransmitters.

Corticospinal fibres from parietal sources end mainly in the contralateral dorsal horn, in the lateral parts of laminae IV–VI and lamina VII. Phylogenetically these fibres represent the oldest part of the corticospinal system. Axons from the sensory cortex terminate chiefly in laminae IV and V. They are concerned with the supraspinal modulation of the transmission of afferent impulses to higher centres, including the motor cortex.

Experimental studies in primates indicate that isolated transection of corticospinal fibres at the level of the pyramid (pyramidotomy) results in flaccid paralysis or paresis of the contralateral limbs and loss of independent hand and finger movements. Destruction of corticospinal fibres at the level of the internal capsule, commonly caused by a cerebral vascular accident or 'stroke', results in a contralateral hemiplegia. The paralysis is initially flaccid but later becomes spastic, and is most marked in the distal muscles of the extremities, especially those concerned with individual movements of the fingers and hand. Associated signs on the paralysed side are: hyperactive deep tendon reflexes; increased tone; the loss of superficial abdominal and cremasteric reflexes; and the appearance of dorsiflexion of the toes (Babinski's sign) in response to stroking the sole of the foot. The latter is usually interpreted as pathognomonic of corticospinal damage, but it is not always present in patients with confirmed corticospinal lesions. Moreover, Babinski's sign is normally present in human infants up to about 2 years of age; its subsequent disappearance may reflect the completion of myelination of the corticospinal fibres and/or the establishment of direct cortical connections to lower motor neurones.

Some of the sequelae of stroke damage in the internal capsule, in particular hyperreflexia and hypertonia, are due to the involvement of other pathways in addition to the corticospinal tract. These include descending cortical fibres to brainstem nuclei, such as the vestibular and reticular nuclei, which also give rise to descending projections that influence motor neurone activity.

Rubrospinal tract

The rubrospinal tract arises from neurones in the caudal magnocellular part of the red nucleus (an ovoid mass of cells situated centrally in the midbrain tegmentum; p. 324). This part of the nucleus contains some 150–200 large neurones, interspersed with smaller neurones.

The origin, localization, termination and functions of rubrospinal connections are poorly defined in humans, and the tract appears to be rudimentary. Rubrospinal fibres cross in the ventral tegmental decussation and descend in the lateral funiculus of the cord, where they lie ventral to, and intermingled with, fibres of the lateral corticospinal tract (see Fig. 20.9). In animals, the tract descends as far as lumbosacral levels, whereas in humans it appears to project only to the upper three

cervical cord segments. Rubrospinal fibres are distributed to the lateral parts of laminae V–VI and the dorsal part of lamina VII of the spinal grey matter. The terminal zones of the tract correspond to those of corticospinal fibres from the motor cortex. Animal studies demonstrate that the effects of rubrospinal fibres on α and γ motor neurones are similar to those of corticospinal fibres.

Tectospinal tract

The tectospinal tract arises from neurones in the intermediate and deep layers of the superior colliculus of the midbrain. It crosses ventral to the periaqueductal grey matter in the dorsal tegmental decussation and descends in the medial part of the ventral funiculus of the spinal cord (see Fig. 20.9). Fibres of the tract project only to the upper cervical cord segments, ending in laminae VI–VIII. They make polysynaptic connections with motor neurones serving muscles in the neck, facilitating those that innervate contralateral muscles and inhibiting those that innervate ipsilateral ones. In animals, unilateral electrical stimulation of the superior colliculus causes turning of the head to the contralateral side, an effect mainly mediated through the tectospinal tract.

Vestibulospinal tracts

The vestibular nuclear complex lies in the lateral part of the floor of the fourth ventricle, at the level of the pontomedullary junction. It gives rise to the lateral and medial vestibulospinal tracts, which are functionally and topographically distinct (Fig. 20.18).

The lateral vestibulospinal tract arises from small and large neurones of the lateral vestibular nucleus (Deiters' nucleus). It descends ipsilaterally, initially in the periphery of the ventrolateral spinal white matter, but subsequently migrating into the medial part of the ventral funiculus at lower spinal levels. Fibres of this tract are somatotopically organized. Thus, fibres projecting to the cervical, thoracic and lumbosacral segments of the cord arise from neurones in the rostroventral, central and dorsocaudal parts, respectively, of the lateral vestibular nucleus. Lateral vestibulospinal fibres end ipsilaterally, mostly in the medial part of the ventral horn in lamina VIII and the medial part of lamina VII.

The medial vestibulospinal tract (see Fig. 20.18) arises mainly from neurones in the medial vestibular nucleus, but some are also located in the inferior and lateral vestibular nuclei. It descends in the medial longitudinal fasciculus into the ventral funiculus of the spinal cord, where it lies close to the midline in the so-called sulcomarginal fasciculus (see Fig. 20.9). Unlike the lateral tract, it contains both crossed and uncrossed fibres, and does not extend beyond the mid-thoracic cord level. Fibres of the medial tract project mainly to the cervical cord segments, ending in lamina VIII and the adjacent dorsal part of lamina VII.

The vestibular nuclei exert a strong excitatory influence upon the antigravity muscles by way of the medial and lateral vestibulospinal tracts. The antigravity muscles include the epaxial muscles of the vertebral column and the extensor muscles of the lower limbs.

Reticulospinal tracts

The reticulospinal tracts pass from the brainstem reticular formation to the spinal cord. The pontine reticulospinal tract (see Fig. 20.15) originates from the pontine tegmentum. The medullary reticulospinal tract arises from the nucleus gigantocellularis, which lies dorsal to the inferior olivary complex. The terminals of reticulospinal fibres are distributed to lamina VIII, and the central and medial parts of lamina VII. The medullary reticulospinal terminals are more widely distributed, ending additionally in the lateral parts of laminae VI and VII, and also directly on motor neurones. Terminations of reticulospinal fibres that originate in the medulla are, in general, more dorsally placed than those that originate in the pons, although there is considerable overlap.

The course and location of the reticulospinal tracts are poorly defined in humans. Some anatomical texts classify the reticulospinal tracts into medial, ventral and ventrolateral tracts, but this is based mainly on studies in animals (see Fig. 20.9). Studies in humans have reported that the reticulospinal fibres in general do not form well-defined tracts but are scattered throughout the ventral and lateral columns (Nathan et al 1996).

Both α and γ motor neurones are influenced by reticulospinal fibres, through polysynaptic and monosynaptic connections. The pontine reticulospinal tract appears to be tonically active and is excitatory to the antigravity muscles, including the epaxial muscles of the vertebral column and the extensor muscles of the lower limbs.

On the other hand, the medullary reticulospinal tract is inhibitory to antigravity muscles. The cells of origin of the medullary reticulospinal tract receive input from the corticospinal tract and the rubrospinal

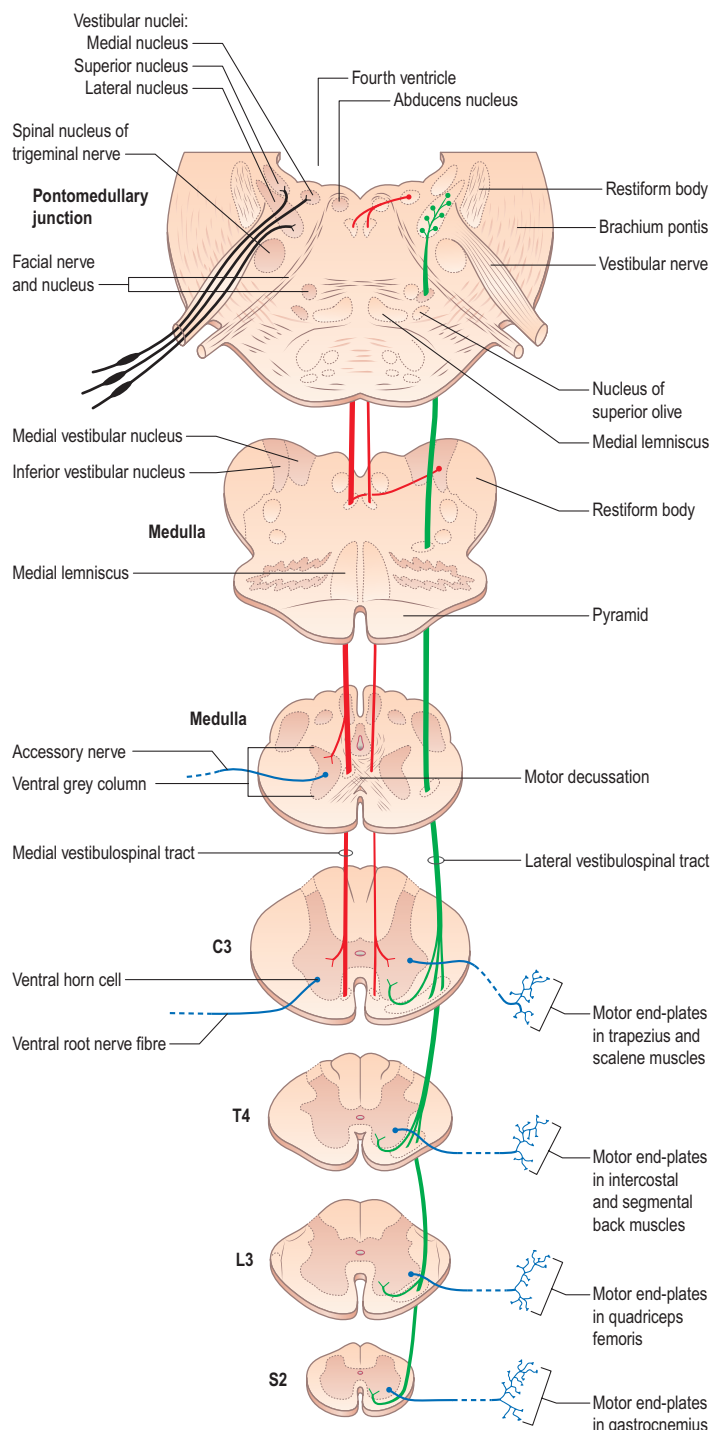


Fig. 20.18 The vestibulospinal tracts. (Redrawn with permission from Carpenter MB 1991 Core Text of Neuroanatomy, 4th edn. Baltimore: Williams and Wilkins.)

tract. The reticulospinal tracts normally function in a coordinated, balanced fashion to control muscle tone. However, if the descending control from higher centres is lost, such as occurs after a stroke in the internal capsule, then the natural excitability of the pontine reticulospinal tract, combined with the activity of the vestibulospinal tracts, causes the antigravity muscles to become hypertonic and hyperreflexic (spasticity) (Guyton and Hall 2006).

Interstitiospinal tract

The interstitiospinal tract arises from neurones in the interstitial nucleus (of Cajal) and the immediate surrounding area, and descends via the medial longitudinal fasciculus into the ventral funiculus of the spinal cord. Its fibres project, mainly ipsilaterally, as far as lumbosacral levels, and are mostly distributed to the dorsal part of lamina VIII and the dorsally adjoining part of lamina VII. They establish some monosynaptic connections with motor neurones supplying neck muscles, but their main connections are disynaptic with motor neurones supplying limb muscles.

Solitariospinal tract

The solitariospinal tract is a small group of mostly crossed fibres that arises from neurones in the ventrolateral part of the nucleus solitarius of the medulla. Descending in the ventral funiculus and ventral part of the lateral funiculus of the cord, these axons terminate on phrenic motor neurones supplying the diaphragm and thoracic motor neurones that innervate intercostal muscles. A pathway with somewhat similar course and terminations to that of the solitariospinal tract originates from the nucleus retroambiguus. Both pathways subserve respiratory activities by driving inspiratory muscles, and some descending axons from the nucleus retroambiguus facilitate expiratory motor neurones. There is clinical evidence that bilateral ventrolateral cordotomy at high cervical levels abolishes rhythmic ventilatory movements.

Hypothalamospinal fibres

Hypothalamospinal fibres exist in animals. They arise from the paraventricular nucleus and other areas of the hypothalamus, and descend ipsilaterally, mainly in the dorsolateral region of the cord, to be distributed to sympathetic and parasympathetic preganglionic neurones in the intermediolateral column. Fibres from the paraventricular nucleus show oxytocin and vasopressin immunoreactivity. They are also distributed to laminae I and X. Descending fibres from the dopaminergic cell group (A11) situated in the caudal hypothalamus innervate sympathetic preganglionic neurones and neurones in the dorsal horn. That similar pathways exist in humans may be inferred from ipsilateral sympathetic deficits (e.g. Horner's syndrome), which follow lesions of the hypothalamus, the lateral tegmental brainstem, or the lateral funiculus of the cord.

Monoaminergic spinal pathways

Monoaminergic cell groups utilize dopamine, adrenaline (epinephrine), noradrenaline (norepinephrine) and 5-HT as neurotransmitters. They occur widely throughout the brainstem and in the hypothalamus. They project rostrally to many forebrain areas and caudally to the spinal cord, and appear to be concerned with the modulation of sensory transmission, and the control of autonomic and somatic motor neuronal activities.

The projections to the spinal cord arise from several sources. Coeruleospinal projections originate from noradrenergic cell groups A4 and A6 in the locus coeruleus complex in the pons and descend via the ventrolateral white matter to innervate all cord segments bilaterally. They end in the dorsal grey matter (laminae IV–VI) and the intermediate and ventral horns. They also project extensively to preganglionic parasympathetic neurones in the sacral cord. Descending noradrenergic fibres, which arise from the lateral tegmental cell groups A5 and A7 of the pons, travel in the dorsolateral white matter. They are distributed to laminae I–III, and particularly to the intermediate grey horn. Descending fibres from adrenergic cell groups C1 and C3 of the medulla oblongata have been traced into the ventral funiculus of the cord and are extensively distributed to the intermediolateral column. Dopaminergic fibres projecting to the spinal cord travel in the hypothalamospinal pathway.

The raphe nuclei pallidus (B1), obscurus (B2) and magnus (B3) in the brainstem give rise to two serotonergic descending bundles. The lateral raphe spinal bundle, from B3 neurones, is concerned with the control of nociception. It descends close to the lateral corticospinal tract and ends in the dorsal horn (laminae I, II and V). The ventral bundle, composed mainly of axons from B1 neurones, travels in the medial part of the ventral white column and ends in the ventral horn (laminae VIII and IX). It facilitates extensor and flexor motor neurones. Some descending serotonergic fibres project to sympathetic preganglionic neurones and are concerned with the central control of cardiovascular function.

Propriospinal pathways

Propriospinal pathways (fasciculi proprii) consist of the ascending and descending fibres of intrinsic spinal neurones. They contact other neurones within the same segment and/or in more distant segments of the spinal cord and so subserve intrasegmental and intersegmental integration and coordination. The majority of spinal neurones are propriospinal neurones, most of which lie in laminae V–VIII. Propriospinal fibres are mainly concentrated around the margins of the grey matter (see Fig. 20.9), but are also dispersed diffusely in the white funiculi.

The propriospinal system plays important roles in spinal functions. Descending pathways end on specific subgroups of propriospinal neurones and these, in turn, relay to motor neurones and other spinal neurones. The system mediates all those automatic functions that

continue after transection of the spinal cord, e.g. sudomotor and vasomotor activities, and bowel and bladder functions.

Some propriospinal axons are very short and span only one segment, while others run the entire length of the cord. The shortest axons lie immediately adjacent to the grey matter and the longer ones are situated more peripherally. Propriospinal neurones can be categorized according to the length of their axons as long, intermediate or short neurones. Long propriospinal neurones distribute their axons throughout the length of the cord, mainly via the ventral and lateral funiculi; their cell bodies are in lamina VIII and the dorsally adjoining part of lamina VII. Axons from the long propriospinal neurones of the cervical cord descend bilaterally, whereas those from the corresponding lumbosacral neurones ascend mainly contralaterally. Most of the fibres are fine (less than 3 μm in diameter). Some are the first spinal tract axons to become myelinated. Intermediate propriospinal neurones occupy the central and medial parts of lamina VII and project mainly ipsilaterally. Short propriospinal neurones are found in the lateral parts of laminae V–VIII and their axons run ipsilaterally in the lateral funiculus.

Propriospinal fibres in the different parts of the white funiculi are distributed preferentially to specific regions of the spinal grey matter. In the spinal enlargements, the propriospinal fibres in the dorsolateral funiculus project to the dorsal and lateral parts of the intermediate zone, and also to spinal motor neurones that supply distal limb muscles, especially those of the hand and the foot. The propriospinal fibres in the ventral part of the ventrolateral funiculus are distributed to the central and medial parts of lamina VII and to motor neurones of proximal limb and girdle muscles. Other propriospinal fibres run in the medial part of the ventral funiculus and travel mainly to the ventromedial part of the intermediate zone, which characteristically contains long propriospinal neurones, and to motor neurones innervating axial and girdle muscles.

The tract of Lissauer

The tract of Lissauer (dorsolateral tract) lies between the apex of the dorsal horn and the surface of the spinal cord, where it surrounds the incoming dorsal root fibres. It is present throughout the spinal cord and is most developed in the upper cervical regions.

The tract consists of fine myelinated and unmyelinated axons. Many are the branches of axons in the lateral bundles of the dorsal roots. These axons bifurcate into ascending and descending branches as they enter the cord. The branches travel in the tract of Lissauer for one or two segments and give off collaterals that end on and around neurones in the dorsal horn. The tract also contains propriospinal fibres, some being short axons of small substantia gelatinosa neurones, which re-enter the dorsal horn.

SPINAL REFLEXES

The intrinsic connections of the spinal cord and brainstem subserve a number of reflexes by which the functions of peripheral structures are modulated in response to afferent information in a relatively automatic or autonomous fashion. The fundamental components of such reflex 'arcs' are an afferent and an efferent neurone, although in all but the simplest of reflexes, interneurones intervene between the afferent and efferent components, conferring increased versatility and complexity on reflex responses. Reflexes may be confined to a single cord level (intra-segmental) or involve several cord segments (inter-segmental). Reflexes, by their very nature, are relatively fixed and stereotyped in form. Nevertheless, they are strongly influenced and modulated by descending connections. In the case of spinal reflexes, these descending controls come from both the brainstem and the cerebral cortex. Pathology of descending supraspinal pathways commonly causes abnormalities of spinal reflex activity, which are routinely tested for in neurological examination; reflexes may be absent (areflexia), diminished (hyporeflexia) or exaggerated (hyperreflexia). During development, descending control mechanisms suppress what may be regarded as 'primitive' spinal reflex responses, such as the extensor plantar reflex and the grasp reflex. When the higher control mechanisms become damaged, these reflexes are released and reappear as a sign of central nervous system pathology (e.g. the Babinski reflex).

Muscle stretch reflex

The muscle stretch reflex (also known as the tendon reflex or deep tendon reflex) is the mechanism by which stretch applied to a muscle elicits its reflex contraction. It is essential for the maintenance of both muscle tone and an upright stance (via the innervation of the postural muscles of the neck, back and lower limbs). Anatomically it is the simplest of reflexes, since it is mediated solely by an afferent and an

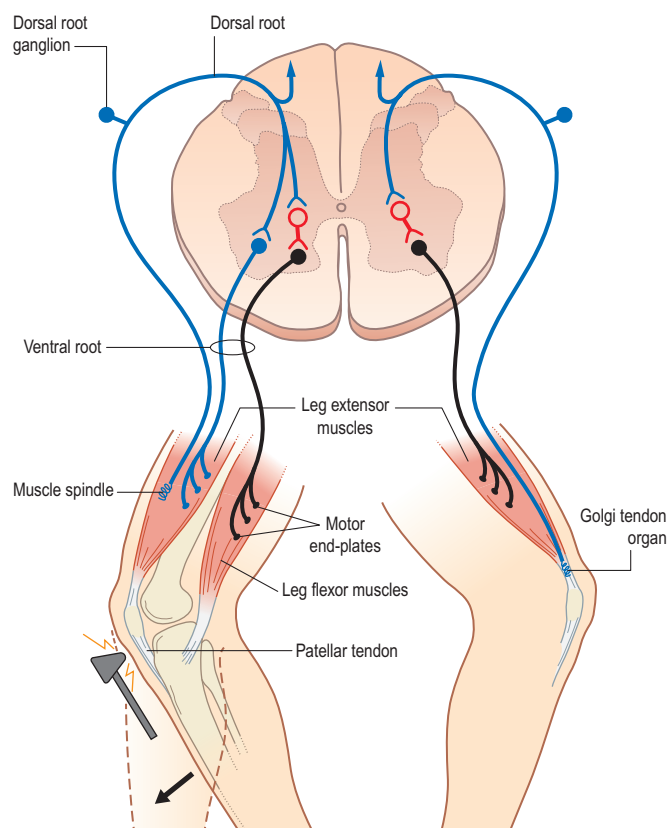


Fig. 20.19 The pathway for the patellar tendon reflex and reciprocal inhibition (left) and autogenic inhibition (right). The inhibitory glycinergic interneurons are represented by the red open cell bodies. (Redrawn with permission from Haines D, *Fundamental Neuroscience for Basic and Clinical Applications*, 4th Edition, 2013, Saunders, Elsevier.)

efferent neurone, and it is therefore also known as a monosynaptic stretch reflex or a myotactic reflex. The afferent component arises from stretch receptors associated with intrafusal muscle fibres located within muscle spindles. The primary or annulospiral endings of these receptive cells give rise to primary afferent fibres that enter the spinal cord, where they make excitatory synaptic contact directly on to α motor neurones innervating the same muscle (Fig. 20.19). The α motor neurones of antagonistic muscles are simultaneously inhibited via collateral connections to inhibitory interneurons (reciprocal innervation).

Gamma reflex

As well as α motor neurones innervating extrafusal muscle fibres, muscles also receive γ motor neurones, which innervate intrafusal muscle fibres. Activation of γ motor neurones increases the sensitivity of the intrafusal fibres to stretch; changes in γ activity have a profound effect on the stretch reflex and on muscle tone. Like α motor neurones, γ motor neurones are under the influence of descending pathways from the brainstem and cerebral cortex. Changes in muscle tone and in the activity of the stretch reflex are commonly found in disorders of the central nervous system as well as the peripheral nervous system.

Flexor reflex and crossed extensor reflex

Painful stimulation of the limbs leads to flexion withdrawal, mediated by a polysynaptic reflex (Fig. 20.20) in which interneurones link the afferent and efferent neurones. Thus, activation of nociceptive primary afferents indirectly causes activation of limb flexor motor neurones. Collateralization of fibres to nearby spinal segments mediates flexion of a limb at several joints, depending on the intensity of the stimulus. Decussating connections to the contralateral side of the cord activate α motor neurones innervating corresponding extensor muscles, which produces a crossed extensor reflex (Fig. 20.21). In principle, virtually any cutaneous stimulus has the potential to induce a flexor reflex but, other than in the case of noxious stimuli, this response is normally inhibited by descending pathways. When descending influences are lost, even harmless cutaneous stimulation can elicit flexion of the limbs. The Babinski (extensor plantar) reflex, which is generally regarded as pathognomonic of damage to the corticospinal tract, is part of a flexion withdrawal of the lower limb in response to stimulation of the sole of the foot.

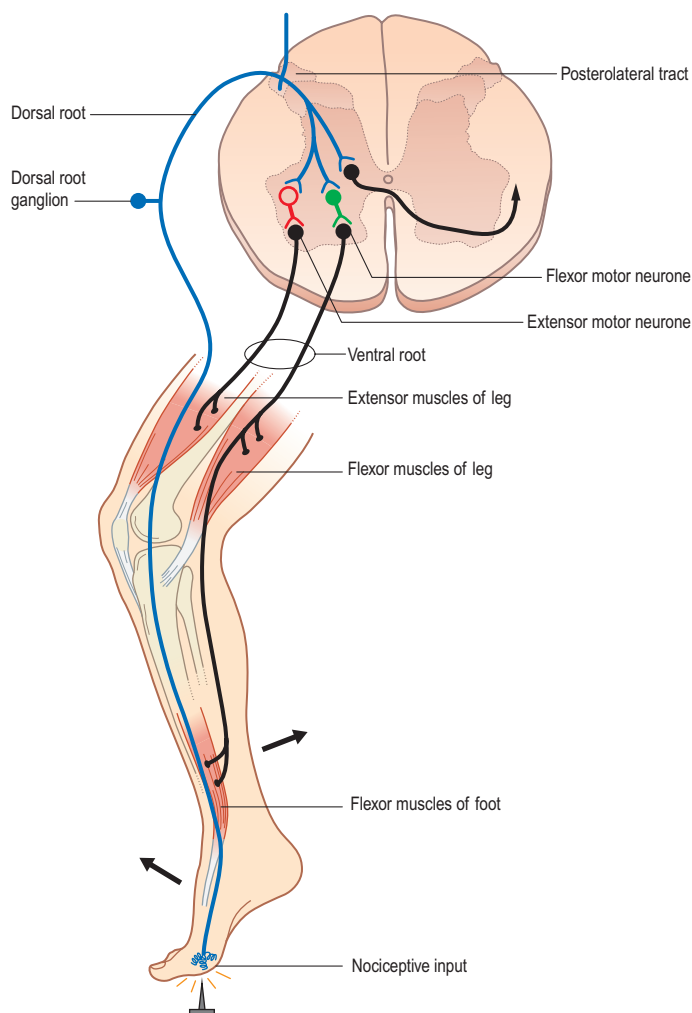


Fig. 20.20 The pathway for the flexor reflex. The inhibitory glycinergic interneurone is represented by the red open cell and the excitatory glutaminergic interneurone by the green closed cell. In addition to being involved in reflexes, this nociceptive input is also relayed to higher levels of the neuraxis via the anterolateral system (ascending black fibre). (With permission from Haines D, *Fundamental Neuroscience for Basic and Clinical Applications*, 4th Edition, 2013, Saunders, Elsevier.)

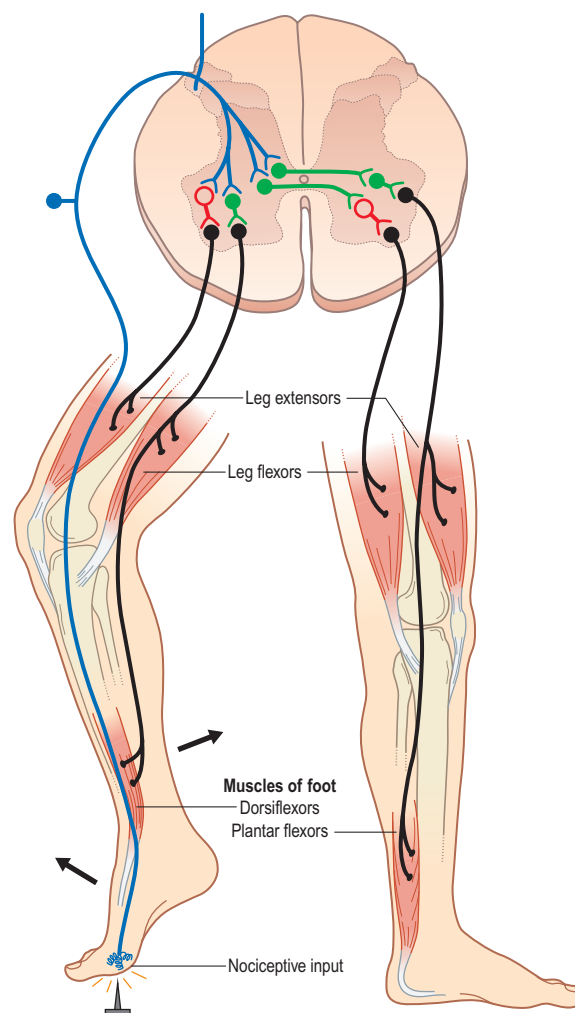


Fig. 20.21 The pathway for the crossed extensor reflex. Glycinergic interneurons (inhibitory) are represented by the open red cell bodies and glutaminergic interneurons (excitatory) by the closed green cells. (Redrawn with permission from Haines D, *Fundamental Neuroscience for Basic and Clinical Applications*, 4th Edition, 2013, Saunders, Elsevier.)

SPINAL CORD LESIONS

Mechanical compression and secondary ischaemic damage to underlying nervous tissue cause surgically relevant spinal cord disease (myelopathy). The site and the level of damage to the cord determine the particular clinical syndrome, e.g. whether the lesion involves the upper or lower cervical, thoracic or lumbosacral spinal cord. At each of these levels, symptoms and signs are determined by direct destruction of segmental tissue, i.e. transversely distributed damage, and disconnection of suprasegmental ascending and descending tracts above and below the level of a lesion, i.e. longitudinally distributed damage (Fig. 20.22). For example, a lower cervical spinal cord lesion damages the segmental sensory and motor contributions to the nerve roots and brachial plexus, causing sensory loss, weakness and wasting of the muscles and loss of tendon reflexes in the upper limbs. Disruption of the ascending sensory pathways in the lateral and dorsal columns of the cervical spinal cord leads to loss of sensation to pain and temperature (lateral spinothalamic tracts) and touch and proprioception (dorsal columns) below the 'sensory level' corresponding to the segment of the spinal cord. Damage to the descending corticospinal tracts in the lateral columns of the spinal cord produces a spastic paraparesis, i.e. increased tone of the muscles, weakness that is of an upper motor neurone pattern (weakness of extensors more than flexors in the upper limbs; weakness of flexors more than extensors in the lower limbs), exaggerated tendon reflexes and abnormal superficial reflexes, e.g. extensor plantar responses

and absent abdominal reflexes. Descending pathways to the bladder are interrupted and this produces a 'neurogenic bladder'.

The same principles apply to lesions at other levels of the spinal cord and they are illustrated in Figure 20.22.

The time course of spinal cord disease is very important in suggesting the likely cause, as the rate of evolution of a disorder varies depending on the aetiology. Disorders with sudden or rapid onset are usually caused by trauma or by a vascular accident (i.e. stroke). When a condition develops over days it is often due to an inflammatory disorder such as multiple sclerosis. Neoplastic lesions such as a spinal cord meningioma often progress over months. Degenerative problems usually progress over months (such as amyotrophic lateral sclerosis or motor neurone disease) or even years (e.g. hereditary spastic paraparesis).

In the online supplementary data there are case histories that illustrate typical patterns of spinal cord disease (Boxes 20.1–20.5). A good knowledge of spinal cord anatomy is vital in making the correct diagnosis.

Bonus e-book image

Fig. 20.5 The basic arrangement of the sensory 'gate' mechanism in the dorsal laminae of the grey matter of the spinal cord.

Box 20.1 Case history 1: Complete transection of the spinal cord

A 29-year-old man fractured his cervical spine after a road traffic accident, leading to transection of the spinal cord at C5. When examined 6 months later, he had paralysis of all four limbs but shoulder abduction and elbow flexion were spared (i.e. all muscles innervated by nerve fibres arising below C5 were paralysed). Tone was increased in the limbs with brisk reflexes and extensor plantar responses (i.e. the weakness was upper motor neurone or pyramidal weakness). He had complete loss of sensation below C5 and a neurogenic bladder.

Box 20.2 Case history 2: Hemisection of the spinal cord giving rise to the Brown–Séquard syndrome

A 20-year-old woman presented with a 4-day history of gradually worsening right leg weakness, burning pain and loss of sensation in the left leg. On examination she had pyramidal pattern weakness of the right leg (i.e. hip and knee flexion and ankle dorsiflexion were mainly affected). Tone was increased in the right leg: there were brisk right leg reflexes and a right extensor plantar response (i.e. right pyramidal weakness). There was impaired proprioception in the right leg. The patient had reduced pinprick sensation affecting the left leg and side up to the umbilicus.

This case illustrates a Brown–Séquard syndrome resulting from hemisection of the cord caused by a right-sided cord lesion in the thoracic spine. Involvement of the pyramidal tract causes pyramidal weakness below the level of the lesion on the same side as the lesion (because the pyramidal tract crosses in the medulla). Involvement of the dorsal columns causes impaired proprioception below the level of the lesion on the same side as the lesion (because the fibres in the dorsal columns also cross in the medulla). Involvement of the spinothalamic tract causes loss of pinprick sensation below the lesion on the contralateral side (because the spinothalamic tract crosses the midline soon after entering the cord). The progression of symptoms over several days suggests an inflammatory problem. Magnetic resonance imaging of the cord showed a right-sided area of cord inflammation (myelitis) at T10. The patient had previously had an episode of inflammation affecting the optic nerve (optic neuritis) and the diagnosis was therefore one of multiple sclerosis.

Box 20.3 Case history 3: Anterior cord syndrome

A 75-year-old man with diabetes and hypertension presented with sudden onset of weakness and sensory disturbance in both legs. When examined 6 months later, he had pyramidal weakness of both legs with increased tone, brisk reflexes and extensor plantar responses, and a complete loss of pinprick sensation below T10. Proprioception was normal.

The acute onset suggested a vascular problem. The diagnosis was one of an anterior spinal artery infarction. Proprioception was spared because the anterior spinal artery supplies only the anterior cord, whereas the posterior cord, including the dorsal columns, is supplied by the posterior spinal arteries, and so was spared.

Box 20.4 Case history 4: Posterior cord syndrome

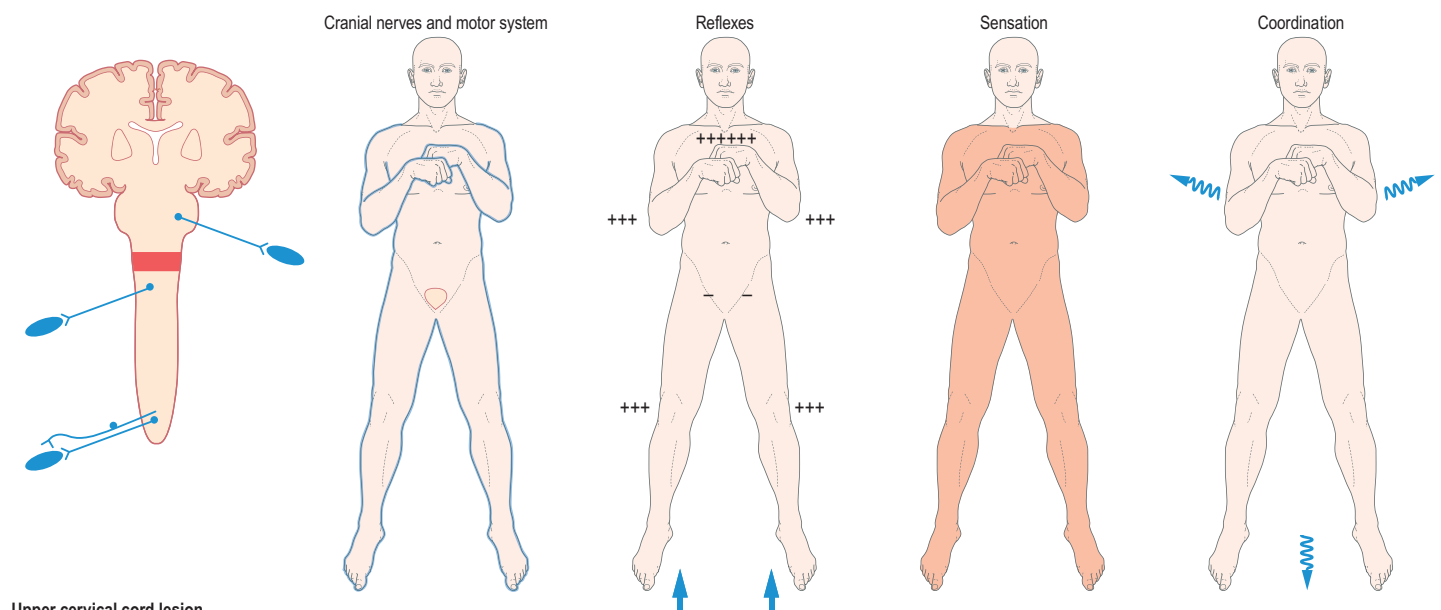
A 73-year-old man presented with a 6-month history of unsteadiness on walking. When examined, he had an unsteady gait, a tendency to fall when asked to stand still with his eyes closed (positive Romberg's test), and impaired proprioception in all four limbs. Spinothalamic sensation (pinprick sensation) was intact and his power was normal.

Investigations revealed signal change in the posterior cervical cord on magnetic resonance imaging and his vitamin B₁₂ levels were very low. A diagnosis of posterior cord syndrome secondary to B₁₂ deficiency was made. Posterior cord syndromes involve the dorsal columns, leading to impaired proprioception. Although not illustrated in this case, B₁₂ deficiency sometimes affects the corticospinal tract, causing weakness in the limbs, and may also sometimes affect the peripheral nerves.

Box 20.5 Case history 5: Central cord syndrome

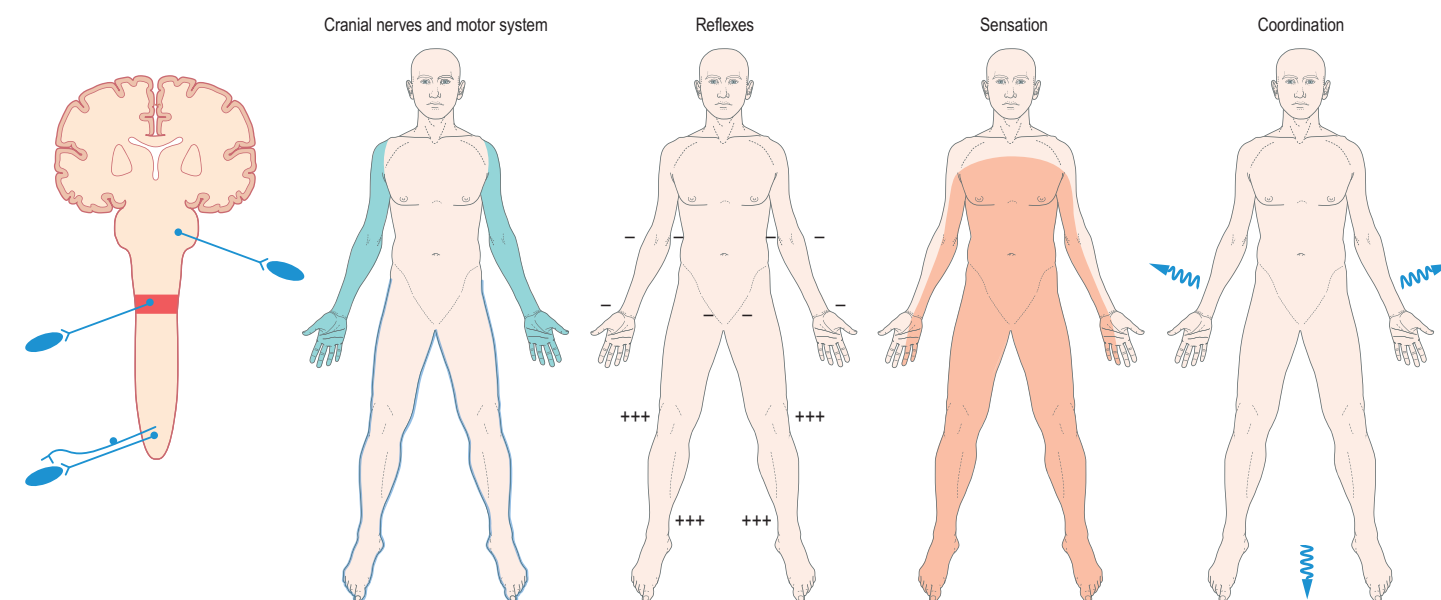
A 20-year-old woman presented with a 4-month history of painless injuries to her hands. For example, after gardening she would sometimes come inside and notice later that there were rose thorns sticking into her hands. Examination showed reduced pinprick sensation (spinothalamic sensation) over the upper limbs.

Magnetic resonance imaging of her cervical spine showed a collection of fluid in the centre of the cervical spinal cord (syringomyelia). A diagnosis of central cord syndrome secondary to syringomyelia was made. As spinothalamic tract fibres cross over from the dorsal horn to run in the contralateral spinothalamic tract, they pass very close to the centre of the cord. Central cord syndromes thus initially damage the spinothalamic tracts, resulting in altered pain sensation at the level of the lesion. As the lesion expands to involve the ventral horns, patients may develop lower motor neurone weakness of the upper limbs. They may subsequently experience upper motor neurone problems in the legs if the lesion expands further to involve the corticospinal tracts.



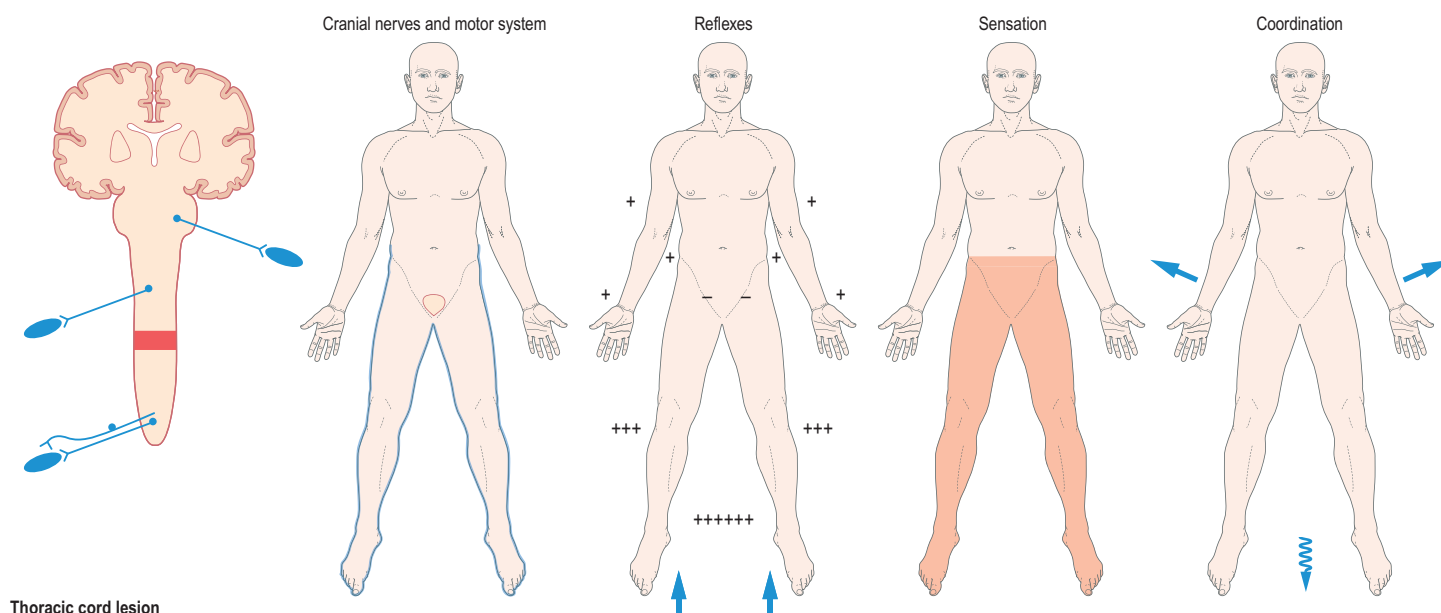
Upper cervical cord lesion

A high cervical cord lesion causes spastic tetraplegia with hyperreflexia, extensor plantar responses (upper motor neurone lesion), incontinence, sensory loss below the level of the lesion and 'sensory' ataxia.



Lower cervical cord lesion

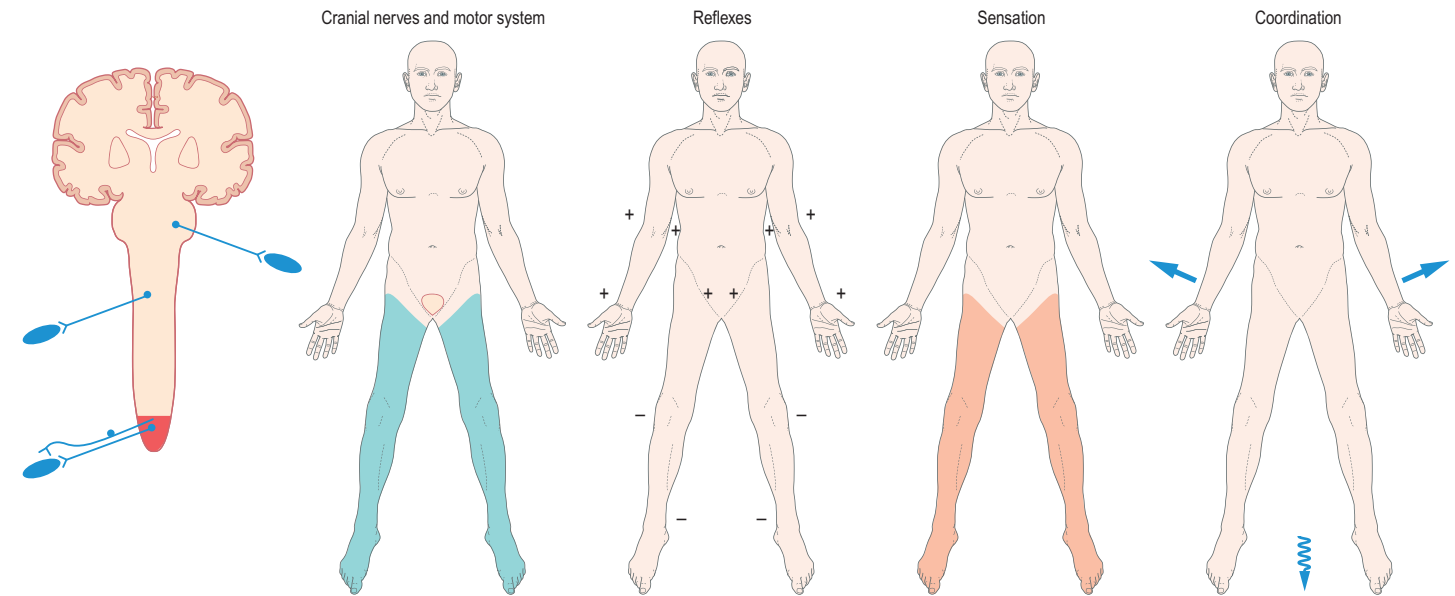
A lower cervical cord lesion causes weakness, wasting and fasciculation of muscles and areflexia of the upper limbs (lower motor neurone lesion). In addition, there is spastic paraparesis, hyperreflexia and extensor plantar responses (upper motor neurone lesion) in the lower limbs, incontinence, sensory loss below the level of the lesion and 'sensory' ataxia.



Thoracic cord lesion

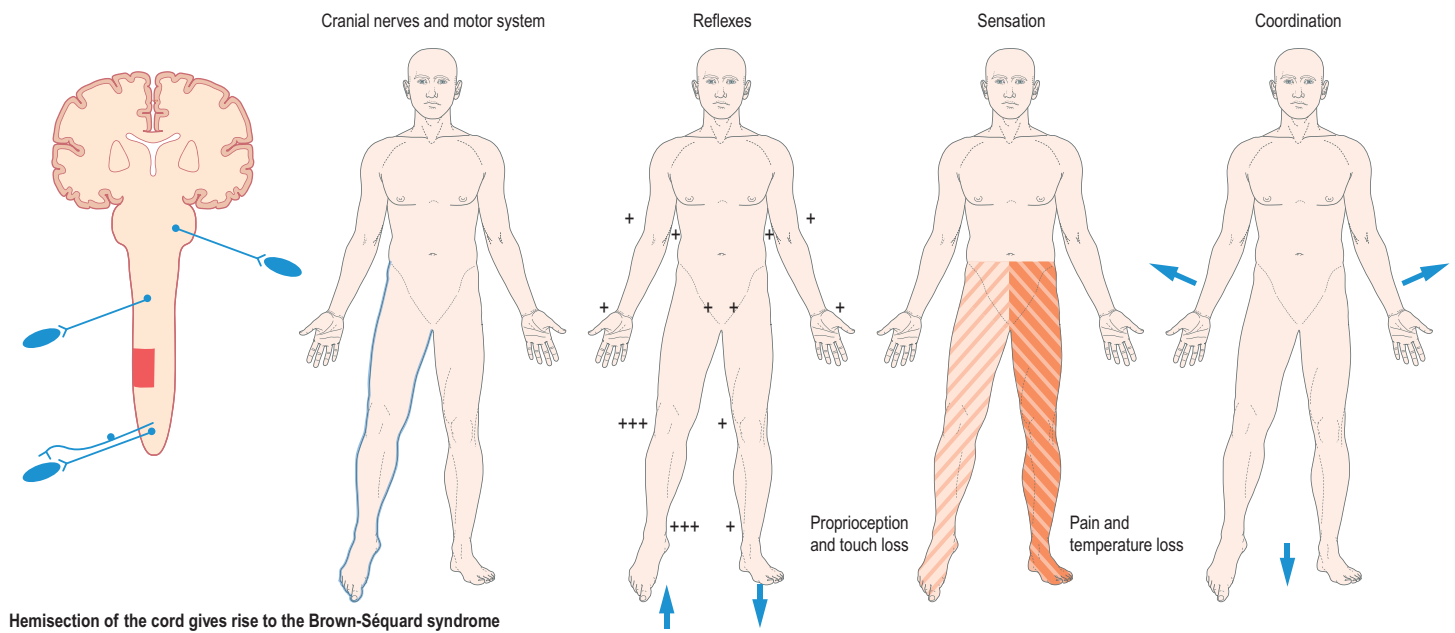
A thoracic cord lesion causes a spastic paraparesis, hyperreflexia and extensor plantar responses (upper motor neurone lesion), incontinence, sensory loss below the level of the lesion and 'sensory' ataxia.

Fig. 20.22 Lesions of the spinal cord. *Continued*



Lumbar cord lesion

A lumbar cord lesion causes weakness, wasting and fasciculation of muscles, areflexia of the lower limbs (lower motor neurone lesion), incontinence, sensory loss below the level of the lesion and 'sensory' ataxia.



Hemisection of the cord gives rise to the Brown-Séquard syndrome

This is characterized by ipsilateral loss of proprioception and upper motor neurone signs (hemiplegia/monoplegia) plus contralateral loss of pain and temperature sensation.

Fig. 20.22, cont'd Lesions of the spinal cord. (With permission from Crossman AR, Neary D 2010 Neuroanatomy, 4th edn. Edinburgh: Churchill Livingstone.)

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